

GABAergic synaptic scaling is triggered by changes in spiking activity rather than AMPA receptor activation

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Abstract

Homeostatic plasticity represents a set of mechanisms that are thought to recover some aspect of neural function. One such mechanism called AMPAergic scaling was thought to be a likely candidate to homeostatically control spiking activity. However, recent findings have forced us to reconsider this idea as several studies suggest AMPAergic scaling is not directly triggered by changes in spiking. Moreover, studies examining homeostatic perturbations *in vivo* have suggested that GABAergic synapses may be more critical in terms of spiking homeostasis. Here we show results that GABAergic scaling can act to homeostatically control spiking levels. We found that perturbations which increased or decreased spiking in cortical cultures triggered multiplicative GABAergic upscaling and downscaling, respectively. In contrast, we found that changes in AMPAR or GABAR transmission only influence GABAergic scaling through their indirect effect on spiking. We propose that GABAergic scaling represents a stronger candidate for spike rate homeostat than AMPAergic scaling.

Significance Statement

The nervous system maintains excitability in order to perform network behaviors when called upon to do so. Networks are thought to maintain spiking levels through homeostatic synaptic scaling, where compensatory multiplicative changes in synaptic strength are observed following alterations in cellular spike rate. Although we demonstrated that AMPAergic synaptic scaling does not appear meet these criteria as a spike rate homeostat, we now show that GABAergic scaling could play this role. Here we present evidence that the characteristics of GABAergic scaling place it in an excellent position to be a spiking homeostat. This work highlights the importance of inhibitory circuitry in the homeostatic control of excitability. Further, it provides a point of focus into neurodevelopmental disorders where excitability is impaired.

eLife assessment

This is an **important** study that brings insight into mechanisms that underlie regulation of GABAergic transmission in response to changes in activity. The authors present **solid** data supporting the premise that action potential firing rather than excitatory synaptic strength is a key determinant of GABAergic synaptic inputs.

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Introduction

Homeostatic plasticity represents a set of compensatory mechanisms that are thought to be engaged by the nervous system in response to cellular or network perturbations, particularly in developing systems ¹. Synaptic scaling is one such mechanism where homeostatic compensations in the strength of the synapses onto a neuron occur following chronic perturbations in spiking activity or neurotransmitter receptor activation (neurotransmission) ². Scaling is typically identified by comparing the distribution of miniature postsynaptic current (mPSC) amplitudes in control and activity-perturbed conditions. For instance, when spiking activity in cortical cultures was reduced for 2 days with the Na⁺ channel blocker TTX or the AMPA/kainate glutamate receptor antagonist CNQX, mEPSC amplitudes were increased ². When first discovered, homeostatic synaptic scaling was thought to be triggered by the cell sensing its reduction in spike rate through reduced calcium entry into the cytoplasm. This was then believed to alter global calcium signaling cascades that led to increased AMPA receptor (AMPA) insertion in a cell-wide manner such that all synapses increased synaptic strength multiplicatively based on each synapse's initial strength ³. In this way excitatory synaptic strength was increased across all of the cell's inputs in order to recover spiking activity without altering relative synaptic strengths resulting from Hebbian plasticity mechanisms. These criteria, sensing spike rate and adjusting synaptic strengths multiplicatively, thus established the expectations for homeostatic synaptic scaling and were consistent with the idea that AMPAergic scaling could be a spike rate homeostat.

More recent work has demonstrated that AMPAergic synaptic scaling is more complicated than originally thought. First, studies have now shown that increases in mEPSC amplitudes or synaptic glutamate receptors often do not follow a simple multiplicative function ^{4,5}. Rather, these studies show that changes in synaptic strength at different synapses exhibit different scaling factors, arguing against a single multiplicative scaling factor that alters synaptic strength globally across the cell. Second, AMPAergic scaling triggered by receptor blockade can induce a synapse-specific plasticity rather than a cell-wide plasticity. Compensatory changes in synaptic strength were observed in several studies where neurotransmission at individual synapses was reduced ⁶⁻⁹. This synapse-specific plasticity would appear to be cell-wide if neurotransmission at all synapses were reduced as occurs in the typical pharmacological blockades that are used to trigger scaling. Regardless, this would still be a synapse specific plasticity, determined at the synapse, rather than the cell sensing its lowered spiking activity through global calcium levels. Finally, several different studies now suggest that reducing spiking levels in neurons is not sufficient to trigger AMPAergic upscaling and therefore bring into question its role as a spike rate homeostat. Forced expression of a hyperpolarizing conductance reduced spiking of individual cells but did not trigger AMPAergic scaling ¹⁰. Further, optogenetic restoration of culture-wide spiking in the presence of AMPAergic transmission blockade triggered AMPAergic scaling that was indistinguishable from that of cultures where AMPAR block reduced spiking (no optogenetic restoration of spiking) ¹¹. Most studies that separate the importance of cellular spiking from

synapse-specific transmission suggest that AMPAergic scaling is triggered by changes in neurotransmission, rather than a cell's spiking activity ^{9,12}. While transmission-dependent AMPAergic scaling appears to be more commonly observed, there are two studies that suggest that alterations in AMPAergic synaptic strength can occur following alterations in spiking in individual cells – AMPA receptor accumulation following blockade of spiking at the soma in cortical cultures ¹³ and reduced mEPSC amplitude following optogenetic activation of individual cells in hippocampal cultures ¹⁴.

Because the pharmacological perturbations that trigger AMPAergic upscaling also result in GABAergic downscaling, it is assumed that they have common triggers. Therefore, in the current study we tested this possibility. Homeostatic regulation of GABAergic miniature postsynaptic current (mIPSC) amplitude was first shown in excitatory neurons following network activity perturbations ¹⁵. Similar to AMPAergic upscaling, chronic perturbations in AMPAR or spiking activity triggered mIPSC downscaling through compensatory changes in the number of synaptic GABA_A receptors ^{15,19}. However, the sensing machinery for triggering GABAergic scaling may be distinct from that of AMPAergic scaling ²⁰. Further, GABAergic plasticity does appear to be a key player in the homeostatic response *in vivo*, as many different studies have shown strong GABAergic compensations following somatosensory, visual, and auditory deprivations ^{21,25}. In addition, these homeostatic GABAergic responses precede and can outlast compensatory changes in the glutamatergic system. Here we describe that GABAergic scaling is triggered by changes in spiking levels rather than changes in AMPAergic or GABAergic neurotransmission, that GABAergic scaling is expressed in a multiplicative manner, and could contribute to the homeostatic recovery of spiking activity. Our results suggest that GABAergic scaling could serve as a homeostat for spiking activity.

Results

TTX and AMPAR blockade triggered a non-uniform scaling of AMPA mPSCs

Previously we have shown that blocking spike activity in neuronal cultures triggered AMPAergic scaling in a non-uniform or divergent manner, such that different synapses scaled with different scaling ratios ^{4,26}. Importantly, these results were consistent across independent studies performed in three different labs using rat or mouse cortical cultures, or mouse hippocampal cultures. We quantitatively evaluated scaling in the following manner. We rank-ordered mEPSC amplitudes (smallest to largest) for both control and TTX-treated cultures and then divided the TTX rank ordered amplitude by the corresponding control rank ordered amplitude (e.g. smallest TTX amplitude divided by smallest control amplitude, etc.) and plotted these ratios for all such comparisons ^{4,26}. Previously, scaling had been thought to be multiplicative, meaning all mPSC amplitudes were altered by a single multiplicative factor. If true for AMPAergic scaling, then our ratio plots should have produced a horizontal line at the scaling ratio. However, we found that ratios progressively increased across at least 75% of the distribution of amplitude ratios. Still, it was unclear whether this was true for all forms of AMPAergic scaling triggered by different forms of activity blockade. Therefore, we repeated this analysis on the data from our previous study ¹¹, but now on AMPAergic scaling produced by blocking AMPAR neurotransmission (40μM CNQX), rather than TTX. We found that the scaling was non-uniform and replicated the scaling triggered by TTX application (Supplemental Figure 1). There was an abrupt increase in the ratio from 1 to ~1.2 (steeper slope) over the first 1-2% of the data, consistent with an error caused by the detection threshold (as shown in simulations of a threshold issue in ⁴). However, ratios then increased over the vast majority of the data from 1.2 to 1.5 more slowly, and this represented the magnitude of homeostatic plasticity with increasing mEPSC amplitude. The results suggest that AMPAergic scaling produced by blocking glutamatergic transmission or spiking in culture was not

multiplicative, but rather different synapses increased by different scaling factors. Further, the similarity of scaling ratio plots following either action potential or AMPAergic blockade is consistent with the idea that they are mediated by similar mechanisms.

TTX and AMPAR blockade reduced both spiking and GABAergic mIPSC amplitude

Previously we made the surprising discovery that AMPAergic upscaling in rat cortical cultures was triggered by a reduction in AMPAR activation rather than a reduction in spiking activity¹¹. Here we tested whether GABAergic scaling was dependent on AMPAR activation or a different trigger, by changes in spiking activity levels. We plated E18 mouse cortical neurons on 64 channel planar multi-electrode arrays (MEAs) and allowed the networks to develop for ~14 days *in vitro* (DIV), a time point where most cultures develop a network bursting behavior (**Figure 1A**, Supplemental Figure 2). We used a custom written Matlab program that was able to detect and compute overall spike rate and burst frequency (Supplemental Figure 2, see methods). We again found that TTX abolished bursts and spiking activity ($n=2$, Supplemental Figure 3). On the other hand, AMPAR blockade (20 μ M) merely reduced bursts and spiking, with a greater effect on bursting. Similar to our findings in rat cortical cultures¹¹, CNQX dramatically reduced burst frequency and maintained this reduction for the entire 24hrs of treatment (**Figure 1B**). Overall spike frequency was also reduced in the first 6 hours, but then recovered over the 24 hour drug treatment (**Figure 1C**). While overall spiking was recovered, we did note that this was highly variable, with some cultures recovering minimally. Following AMPAergic blockade bursts continued in these cultures, likely due to NMDAergic neurotransmission as shown previously¹¹.

In order to examine the possibility that compensatory changes in GABAergic synaptic strength could have contributed to the recovery of the network spiking activity we assessed synaptic scaling by measuring mIPSC amplitudes in pyramidal-like neurons in a separate set of cortical cultures plated on coverslips. We found that both activity blockade with TTX, and AMPAergic blockade with CNQX, triggered a dramatic compensatory reduction in mIPSC amplitude compared to control (untreated) cultures (**Figure 2A**). Even though TTX completely abolished spiking, while CNQX only reduced spiking, both treatments triggered a similar reduction in average mIPSC amplitude. In order to more carefully compare the GABAergic scaling that is triggered by TTX and CNQX mechanistically, we created scaling ratio plots as described above⁴. In **Figure 2B** we show that TTX-induced and CNQX-induced scaling does produce a largely multiplicative downscaling with a scaling factor around 0.5. This is consistent with the idea that the mechanisms of GABAergic scaling were similar following activity or AMPAergic blockade. We noticed that the first mIPSC ratios started near one and within 50 ratios came down to the 0.5 level (**Figure 2B**). This is likely due to the smallest drug-treated mIPSCs falling below our detection cutoff of 5 pAs⁴. On the other hand, the largest mIPSCs trended above 0.5, consistent with the possibility that a small portion of the mIPSCs may not scale uniformly. Together, these results are consistent with the idea that either spiking or reduced AMPA receptor activation could trigger the GABAergic downscaling since TTX or CNQX would reduce both.

Optogenetic restoration of spiking in the presence of AMPAR blockade prevented GABAergic downscaling

In order to separate the importance of spiking levels from AMPAR activation in triggering GABAergic downscaling we blocked AMPARs while restoring spike frequency. Cultures were plated on the MEA and infected with ChR2 under the human synapsin promoter on DIV 1. Experiments were carried out on ~ DIV14, when cultures typically express network bursting. Baseline levels of spike frequency were measured in a 3-hour period before the addition of 20 μ M CNQX (**Figure 3A**). We then used a custom written TDT Synapse software that triggered a brief (50-100ms) activation of a blue light photodiode to initiate bursts (see methods, **Figure 3B**) whenever the

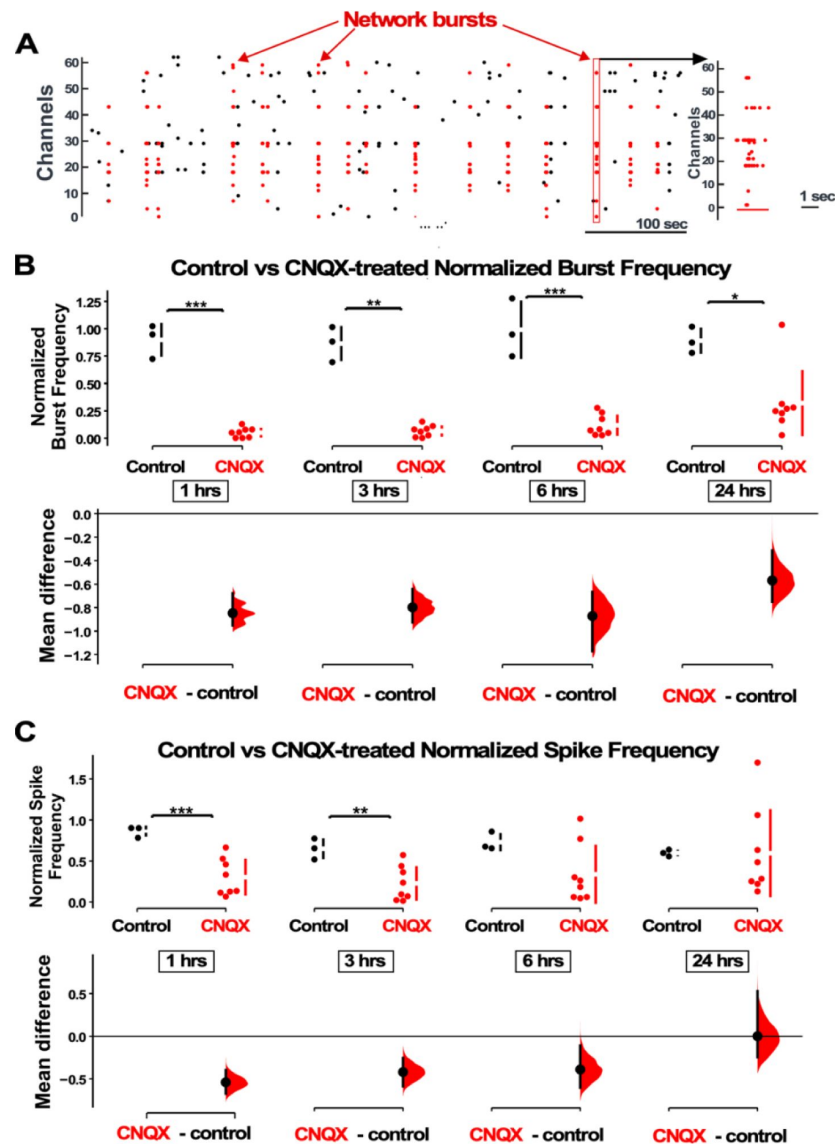


Figure 1.

AMPAergic blockade reduces burst frequency and overall spike rate. **A**) Network bursts can be identified by detected spikes (red dots) time-locked in multiple channels of the MEA (Y axis). One burst (red rectangle) is expanded in time and shown in the raster plot on the right. **B**) The normalized burst rate is shown in control cultures and following application of CNQX for 24 hrs. **C**) Average overall spike frequency is compared for CNQX-treated cultures and control unstimulated cultures at 1hr, 3hrs, 6hrs ($p=0.104$), and 24hrs ($p=0.982$) after addition of CNQX or vehicle. The mean differences at different time points are compared to control and displayed in Cumming estimation plots. Significant differences denoted by * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$. Recordings from single cultures (filled circles, control $n = 3$ cultures, CNQX $n = 8$ cultures), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. Mean differences between control and treated groups are plotted on the bottom panel, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars).

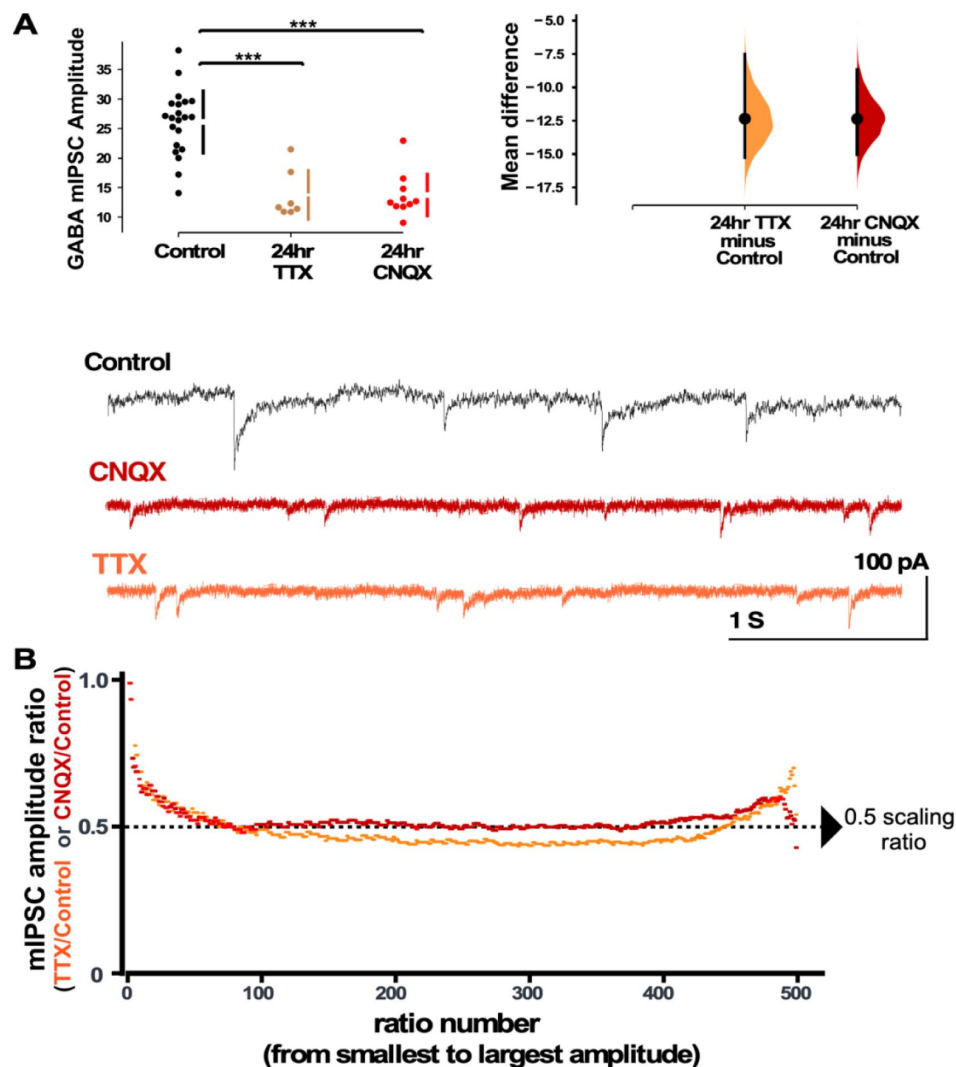


Figure 2.

Both activity and AMPAR blockade cause a reduction in mIPSC amplitudes that appear to scale down. **A**) CNQX and TTX produce a reduction in average amplitude of mIPSCs as shown in the scatter plots (control - $n=21$ from 10 cultures, TTX - $n=7$ from 3 cultures, CNQX - $n=10$ from 6 cultures). The mean differences are compared to control and displayed in Cumming estimation plots. Significant differences denoted by *** $p \leq 0.001$. GABAergic mIPSC amplitudes from single neurons (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the panels to the left. Mean differences between control and treated groups are plotted on the panel to the right, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars). Example traces showing mIPSCs are shown below. **B**) Scaling ratio plots show the relationship of mIPSC amplitudes from treated cultures compared to untreated cultures. All recordings taken from cultured neurons plated on coverslips, not MEAs.

running average of the firing rate fell below the baseline level, established before the addition of the drug. In this way we could optically initiate bursts that largely occurred after the blue light was off. These optically-induced bursts look very similar to the spontaneously occurring pre-drug bursts and this largely restored the spike rate to pre-drug values (**Figure 3B**). We used 20 μ M CNQX to block AMPARs, instead of the 40 μ M concentration that we used in the previous study¹¹ because 40 μ M CNQX severely impaired our ability to optogenetically restore spiking activity in these cultures.

We have already established that bursts and spiking were reduced following the application of CNQX (**Figure 1**). However, when we optogenetically activated the cultures in the presence of CNQX we found that both the burst rate and spike frequency were increased compared to CNQX treatment alone, no optostimulation (Supplemental Figure 4). Because the program was designed to maintain total spike frequency, photostimulation of CNQX-treated cultures did a relatively good job at recovering this parameter to control levels (**Figure 3D**). In fact, spike frequency was slightly, but not significantly, above control levels through the 24 hour recording period (**Figure 3D**). In our previous study we were able to establish that the optogenetically-evoked bursts in CNQX and even the pattern of individual unit spiking during the burst was restored to that of normally occurring bursts in the pre-drug condition¹¹. On the other hand, the program designed to control overall spike frequency through optostimulation in CNQX did not completely return burst frequency back to control levels (**Figure 3C**).

We next assessed mIPSC amplitudes using whole cell recordings taken from cultures plated on MEAs. After blocking AMPAR activation without optogenetic restoration of spiking activity, we found that mIPSC amplitudes were significantly reduced compared to control conditions (**Figure 4A**), as we had shown for CNQX treatment on cultures plated on coverslips (**Figure 2A**). Strikingly, when spiking activity was optogenetically restored in the presence of CNQX for 24 hours we observed that mIPSCs were no different than control values (same as control, larger than CNQX only – **Figure 4A**). This result suggested that unlike AMPAergic upscaling, GABAergic downscaling was prevented if spiking activity levels were restored in the presence of AMPAR blockade. In order to compare scaling profiles we plotted the scaling ratios for these different treatments. Not surprisingly, we found that MEA-plated cultures treated with CNQX but given no optogenetic stimulation were similar to CNQX-treated cultures plated on coverslips (CNQX/control ~ 0.5 , **Figure 4B** vs **Figure 2B**). Ratio plots of cultures treated with CNQX where activity was restored optogenetically compared to controls demonstrated a fairly uniform relationship with a ratio of around 1 through most of the distribution suggesting the mIPSCs in these two conditions were similar and therefore unscaled (**Figure 4B**). Interestingly, the scaling ratio and the average mIPSC amplitudes in the optogenetically activated cultures were slightly larger than control mIPSCs which may be due to the slight increase in spiking in optogenetically stimulated cultures. Together, these results suggest that GABAergic downscaling was triggered by reductions in spiking activity, independent of AMPA receptor activation, and was multiplicative in that the vast majority of mEPSC amplitudes ($\sim 95\%$) appeared to be reduced to $\sim 50\%$.

Enhancement of AMPAR currents triggered GABAergic upscaling in a spike-dependent manner

While reductions in spiking activity triggered a GABAergic downscaling, it was less clear whether increases in spiking activity could trigger compensatory GABAergic upscaling. To test for such a possibility, we exposed the cultures to cyclothiazide (CTZ), an allosteric enhancer of AMPA receptors that also enhances spontaneous glutamate release¹¹. Due to the hydrophobic nature of CTZ it was necessary to dissolve it in ethanol, and used ethanol without CTZ as a control (final solution 1:1000 ethanol in Neurobasal). In addition to increasing AMPAR activation, CTZ application slightly increased overall spiking activity in our MEA-plated cultures in the first 3 hours of the drug, although this was quite variable (**Figure 5A-B**). The amplitude of mIPSCs in

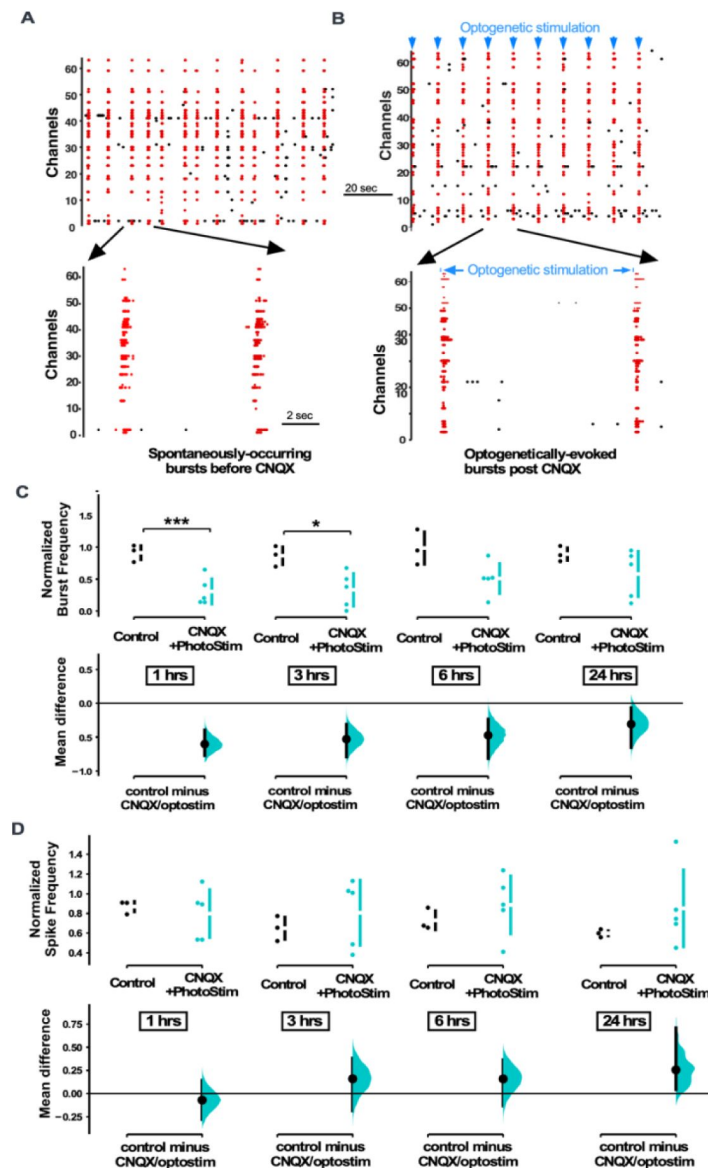


Figure 3.

MEA recordings show that optogenetic stimulation restores spiking activity in cultures treated with CNQX. **A**) Spontaneously-occurring bursts of spiking are identified (synchronous spikes/red dots). Expanded version of raster plot highlighting 2 bursts is shown below. **B**) Same as in A, but after CNQX was added to the bath and bursts were now triggered by optogenetic stimulation (blue line shows duration of optogenetic stimulation). **C**) Average burst rate is compared for CNQX-treated cultures with optogenetic stimulation (n=5 cultures) and control unstimulated cultures (n=3 cultures) at 1 hr, 3 hrs, 6 hrs ($p=0.056$), and 24 hrs ($p=0.379$) after addition of CNQX or vehicle (same control data presented in [Figure 1](#)). **D**) Average overall spike frequency is compared for CNQX-treated cultures with optogenetic stimulation and control unstimulated cultures at 1 hr ($p=0.612$), 3 hrs ($p=0.489$), 6 hrs ($p=0.449$), and 24 hrs ($p=0.22$) after addition of CNQX or vehicle. Control data is same as presented in [Figure 1](#). The mean differences at different time points are compared to control and displayed in Cumming estimation plots. Significant differences denoted by * $p \leq 0.05$, *** $p \leq 0.001$. Recordings from single cultures (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. Mean differences between control and treated groups are plotted on the bottom panel, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars).

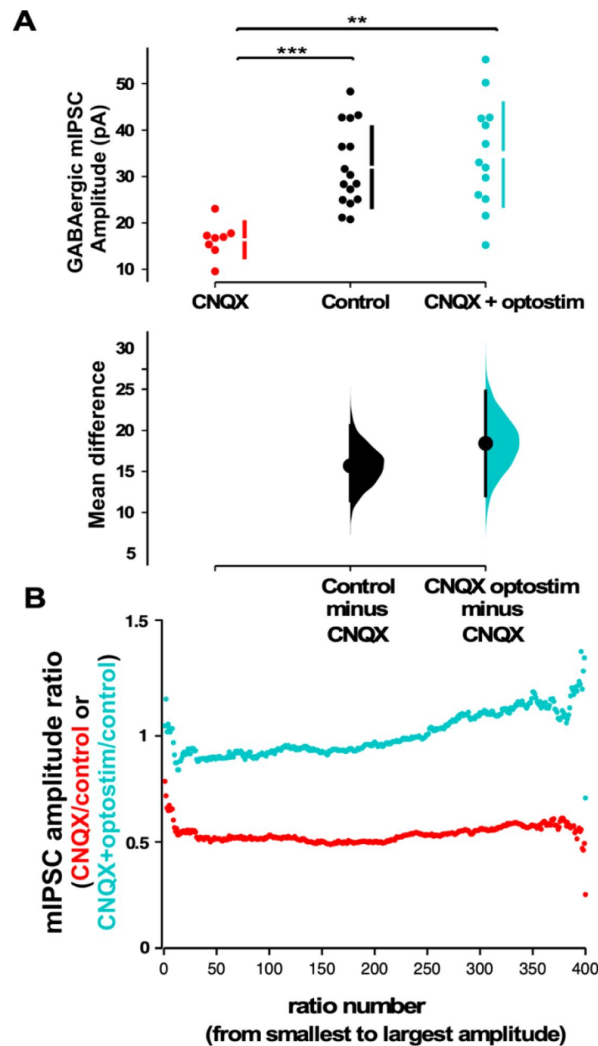


Figure 4.

Optogenetic restoration of spiking activity in the presence of AMPAR blockade prevents GABAergic downscaling observed in CNQX alone. **A**) Scatter plots show AMPAR blockade triggers a reduction in mIPSC amplitude compared to controls that is prevented when combined with optogenetic stimulation (optostim, control - $n=16$ from 10 cultures, CNQX - $n=8$ from 4 cultures, CNQX/optostim - $n=13$ from 6 cultures). The mean differences are compared to control and displayed in Cumming estimation plots. Significant differences denoted by ** $p \leq 0.01$, *** $p \leq 0.001$. GABAergic mIPSC amplitudes from single neurons (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. Mean differences between control and treated groups are plotted on the bottom panel, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars). **B**) Scaling ratio plots show largely multiplicative relationships to control values for both CNQX and CNQX + photostimulation treatments. Cultured neurons for these recordings were obtained from cells plated on MEAs (control, CNQX, and CNQX+optostim).

control cultures exposed to ethanol were no different than control cultures without ethanol (**Figure 5C**). We then treated coverslip-plated cultures with CTZ for 24 hours and found that this did indeed produce a compensatory increase in GABA mIPSC amplitude (**Figure 5D**). In our previous study we found that enhancing AMPAergic neurotransmission in the presence of activity blockade (CTZ + TTX) reduced AMPAergic upscaling compared to activity blockade alone (TTX) ¹¹. Therefore, we tested whether enhancing AMPAergic neurotransmission in activity blockade (CTZ + TTX) altered GABAergic scaling induced by TTX alone (24 hrs). Here we found that the GABAergic downscaling following TTX was no different when AMPAergic neurotransmission was enhanced (CTZ + TTX, **Figure 5D**). To determine if these changes in mIPSC amplitude were of a multiplicative scaling nature we made ratio plots. This demonstrated that both CTZ increases and CTZ+TTX decreases in mIPSC amplitude were multiplicative and therefore represented scaling (**Figure 5E**, CTZ – scaling ratio of 1.5, CTZ+TTX - scaling ratio of 0.6). Further, the scaling ratio plot for CTZ + TTX looked similar to those of TTX alone (compare **Figure 5E** and **2B**). These results showed a compensatory upward and downward GABAergic scaling, that more closely followed spiking activity compared to AMPAergic transmission.

Blocking GABAergic receptors for 24 hours triggered upscaling of GABAergic mIPSCs

The above results suggested that GABAergic scaling was more dependent on the levels of spiking activity. However, one alternative possibility was that these changes in GABA mIPSCs were due to changes in GABAergic receptor activation, which could act as a proxy for activity levels. In this way, GABARs sense changes in spiking activity levels and directly trigger GABAergic scaling to recover activity. To address this possibility, we treated cultures with the GABA_A receptor antagonist bicuculline to chronically block GABAergic receptor activation while increasing spiking activity. If increased spiking activity is directly the trigger (not mediated through GABAR activity), then we would expect to see GABAergic upscaling. On the other hand, if GABAR activation is a proxy for spiking then blockade of these receptors would indicate low activity levels and we would expect a downscaling to recover the apparent loss of spiking. GABAR block produced an upward trend in both burst frequency (**Figure 6A**) and spike frequency (**Figure 6B**). We measured mIPSCs in a separate cohort of cultures plated on coverslips which were treated with bicuculline for 24 hours, and we observed GABAergic upscaling (**Figure 6C**). These results are consistent with previous work in hippocampal cultures that showed GABAergic upscaling following bicuculline treatment^{18, 28}. We also assessed mIPSC frequency in all of the drug conditions but did not observe significant differences, possibly due to the tremendous variability of this feature (Supplemental Figure 5). Our results were consistent with the idea that direct changes in spiking activity, rather than AMPA or GABA receptor activation triggered compensatory GABAergic upscaling. The scaling ratio plots were again relatively flat, with a scaling ratio of around 1.5, suggesting a multiplicative GABAergic upscaling (**Figure 6D**) that was similar to CTZ-induced upward scaling (**Figure 5E**).

The trigger for GABAergic and AMPAergic scaling is distinct in mouse cortical cultures

We have shown the importance of alterations of spiking activity in triggering GABAergic scaling in mouse cortical cultures. Previously, we had shown that AMPAergic scaling was dependent on glutamatergic transmission rather than spiking, and did this in rat cortical cultures. This is a striking result as we had expected these homeostatic mechanisms to share a common trigger. To ensure that the triggers for AMPAergic and GABAergic scaling really were distinct in the same culture set and conditions used in the present study, we repeated our experiment blocking AMPAR activation for 24 hrs with 20μM CNQX, but now checked for AMPAergic scaling. We found the surprising result that following 24 hr CNQX treatment there was no change in AMPAergic mEPSC

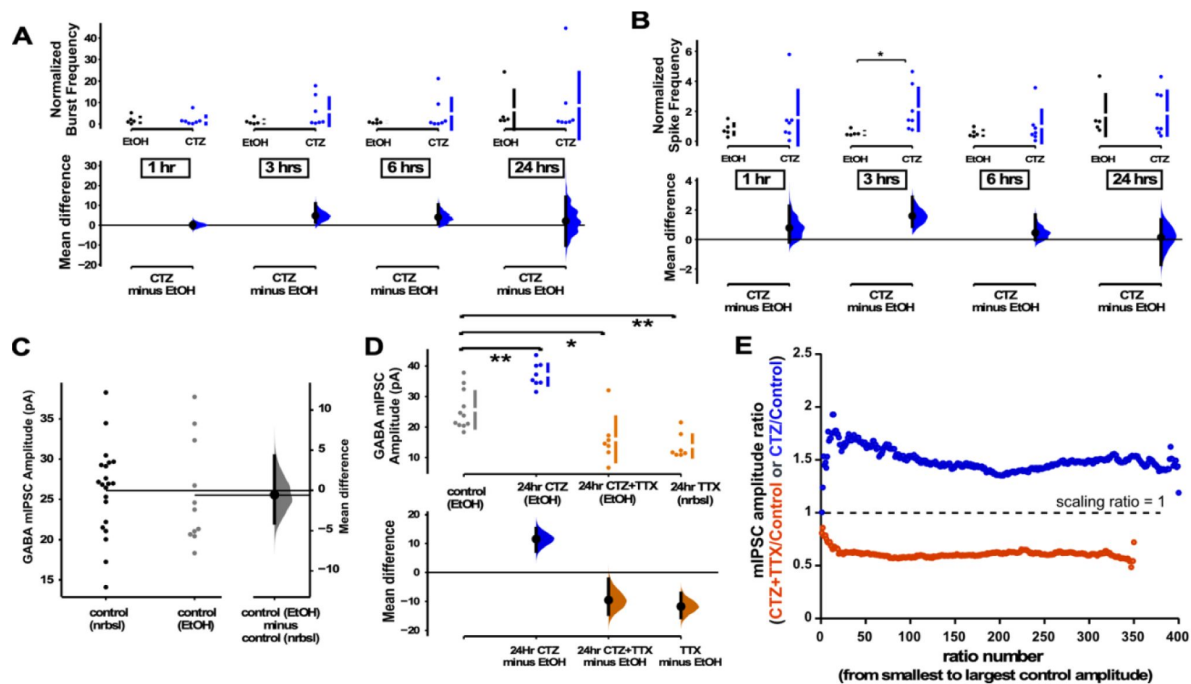


Figure 5.

GABAergic upscaling was also triggered by CTZ and this was dependent on spiking activity. **A)** MEA recordings show that CTZ-treated cultures trended toward increases in normalized burst rate compared to control untreated cultures at 1hr ($p=0.97$), 3hrs ($p=0.246$), 6hrs ($p=0.397$), and 24hrs ($p=0.894$) after addition of CNQX ($n=7$) or vehicle ($n=5$). **B)** MEA recordings show that CTZ-treated cultures trended toward increases in normalized overall spike rate compared to control untreated cultures at 1hr ($p=0.565$), 3hrs, 6hrs ($p=0.634$), and 24hrs ($p=0.92$) after addition of CNQX or vehicle. **C)** Control cultures in Neurobasal (nrbsl) were compared with control cultures with ethanol (EtOH) dissolved in Neurobasal (1:1000). Amplitude of mIPSCs in different controls were no different ($p=0.803$, nrbsl - $n=21$ from 10 cultures, EtOH - $n=11$ from 3 cultures). **D)** CTZ treatment (dissolved in ethanol) led to an increase in mIPSC amplitude compared to ethanol control cultures (CTZ - $n=8$ from 3 cultures). CTZ combined with TTX (in ethanol) produced a reduction of mIPSC amplitude compared to controls (ethanol) that was no different than TTX (nrbsl) alone (CTZ+TTX - $n=7$ from 3 cultures, TTX - $n=7$ from 3 cultures is same data as shown in **Figure 2A**). The mean differences at different time points or conditions are compared to control and displayed in Cumming estimation plots. Significant differences denoted by * $p \leq 0.05$, ** $p \leq 0.01$. Recordings from single cultures (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. Mean differences between control and treated groups are plotted on the bottom panel, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars). **E)** Scaling ratios show that both CTZ-induced increases and CTZ+TTX-induced decreases were multiplicative. All mIPSC amplitudes recorded from cultures plated on coverslips, not MEAs.

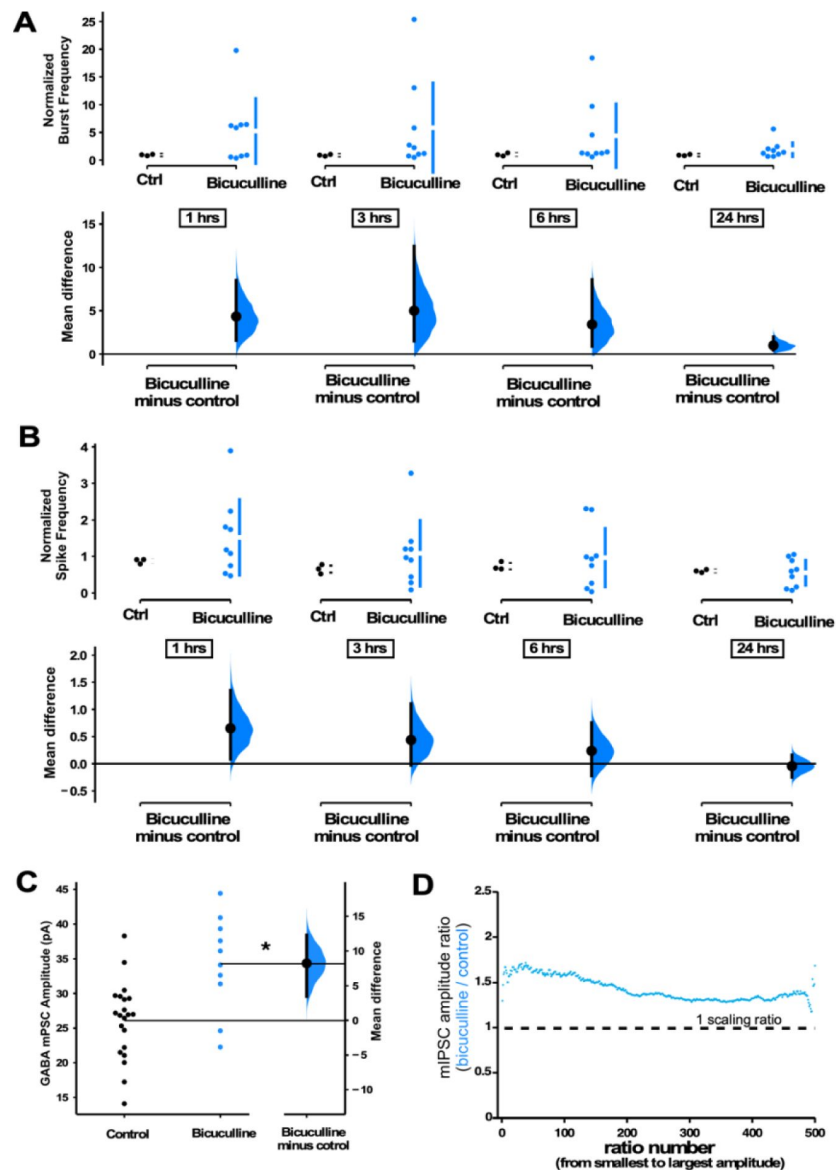


Figure 6.

GABAergic upscaling is triggered by increased spiking activity rather than reduced GABAR activation. **A**) Bicuculline-treated cultures (24hrs) plated on MEA's trended upward in normalized burst rate compared to control untreated cultures at 1hr ($p=0.314$), 3hrs ($p=0.246$), 6hrs ($p=0.339$), and 24hrs ($p=0.339$) after addition of bicuculline ($n=9$ cultures) or vehicle ($n=3$ cultures, same data as [Figure 1](#)). **B**) Bicuculline-treated cultures (24hrs) plated on MEA's trended upward in normalized overall spike frequency compared to control untreated cultures at 1hr ($p=0.358$), 3hrs ($p=0.462$), 6hrs ($p=0.734$), and 24hrs ($p=0.772$) after addition of bicuculline or vehicle. Recordings from single cultures (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. **C**) Bicuculline treatment (24hrs) produced an increase in mIPSC amplitudes (control $n=21$ from 10 cultures, bicuculline - $n=10$ from 4 cultures). The mean difference is compared to control and displayed in Cumming estimation plots. Significant difference denoted by * $p \leq 0.05$. Recordings from single neurons (filled circles), and mean values (represented by the horizontal line). Control and treated group is plotted, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CI is depicted by vertical error bar). **D**) Ratio plots for bicuculline-induced increase in mIPSCs exhibits a multiplicative profile. All mIPSC amplitudes recorded from cultures plated on coverslips, not MEAs.

amplitudes (**Figure 7**), despite the fact that this was the same treatment that reduced spiking activity in our cultures and triggered GABAergic downscaling. The result confirms the observation that the triggers for AMPAergic and GABAergic scaling in the same cultures are distinct.

Discussion

In the original study describing AMPAergic synaptic scaling, the authors triggered this plasticity by blocking spiking activity with TTX or blocking AMPAergic neurotransmission with CNQX². Similar results have now been demonstrated in multiple tissues and labs²⁶. It was thought that AMPAergic scaling was a homeostatic mechanism, triggered by alterations in spiking and likely calcium transients associated with cellular spiking; once the cell drifted outside the setpoint for spiking a cell-wide signal was activated that changed the synaptic strengths of all AMPAergic inputs by a single multiplicative scaling factor to return the cell to the spiking set point³. In this way, AMPAergic scaling could homeostatically regulate spiking levels, while also preserving the relative differences in synaptic strength that have been set up by Hebbian plasticity mechanisms. However, as described in the introduction, more recent studies suggest that AMPAergic synaptic scaling does not appear to meet these initial expectations.

Previous work suggests AMPAergic scaling following TTX or TTX+APV treatment was not multiplicative^{4,5}, and we now show that it is not multiplicative following AMPAR blockade (CNQX treatment, Supplemental Figure 1). Further, several studies suggest that changes in mEPSC amplitude associated with AMPAergic scaling occur at the level of the synapse rather than globally throughout the cell. In fact, several studies have suggested that glutamate receptor activation due to action potential-independent spontaneous release could play a significant role in triggering AMPAergic scaling^{7,11,29}. It is certainly possible that there are compensatory changes in mEPSC amplitude that can be triggered by either altered neurotransmission or spiking. However, as we have shown previously, putting back significant spiking activity levels and their associated calcium transients in the presence of CNQX had no effect on AMPAergic scaling (no reduction in the existing scaling¹¹). Because AMPAergic scaling does not directly follow spiking activity levels, it does not appear to fulfill the expectations of a homeostat for spiking. Rather, AMPAergic scaling in many cases seems to act to homeostatically maintain the effectiveness of individual synapses.

GABAergic scaling appears to exhibit all the features initially predicted for AMPAergic synaptic scaling. First, GABAergic scaling is multiplicative, meaning the relative strengths of these synapses can be maintained (**Figures 2**, 4-6). Critically, GABAergic scaling can act as a firing rate homeostat for the following reasons. GABAergic downscaling was triggered by alterations in spike rate, rather than AMPAergic neurotransmission. We found that CNQX-triggered GABAergic downscaling was abolished when we optogenetically restored spiking activity levels (**Figure 3-4**), and that increasing spiking with bicuculline or CTZ both triggered GABAergic upscaling (**Figures 5-6**). While we cannot rule out a role of AMPAR activation in GABAergic upscaling, we did observe that CTZ-induced upscaling was converted to downscaling in the presence of TTX (**Figure 5C-D**). Further, the findings suggest that altering neurotransmission did not contribute to GABAergic scaling. Increasing AMPAergic transmission with CTZ in the presence of TTX had no impact on downscaling as it was no different than following TTX treatment alone (**Figure 5D**). Also, if GABAR transmission were a proxy for activity levels, then blocking GABA_A receptors would mimic activity blockade and should lead to a compensatory downscaling. However, bicuculline (reduced GABAR activity) increased spiking and triggered a GABAergic upscaling consistent with the idea that spiking was the critical feature (**Figure 6**). This result was consistent with previous work in hippocampal cultures where chronic bicuculline treatment triggered GABAergic upscaling, which was prevented if the cell was hyperpolarized¹⁸. Finally, if scaling contributed to a homeostatic recovery of activity, then GABAergic scaling should have been expressed by 24 hours of CNQX (before bursts and before spike frequency in some cultures had fully recovered,

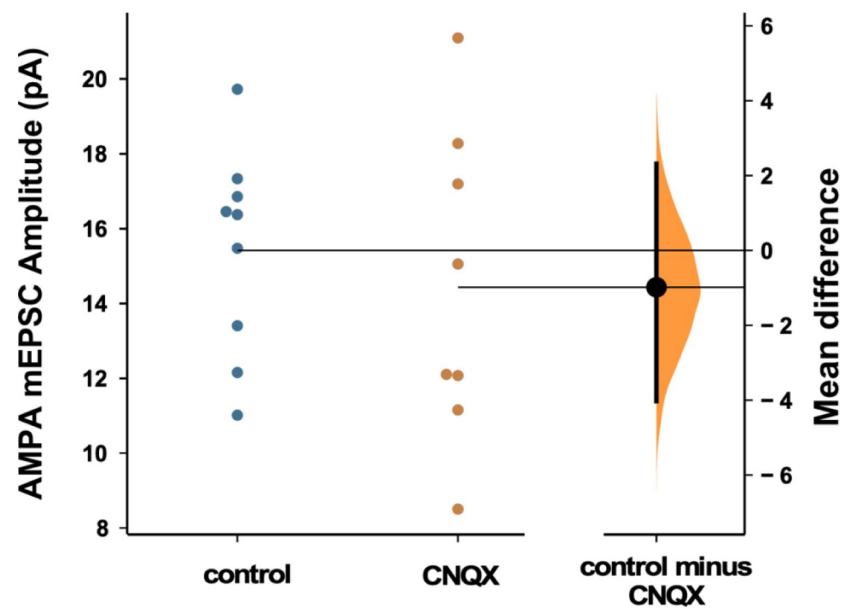


Figure 7.

AMPAergic scaling was absent following 24 hours of 20 μ M CNQX. AMPA mEPSC amplitudes were no different than control following AMPAR blockade ($p=0.57$, control - $n=9$ from 4 cultures, CNQX - $n=8$ from 3 cultures). Recordings from single neurons (filled circles), where mean values (represented by horizontal bar) are plotted, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars). All mEPSC amplitudes recorded from cultures plated on MEAs.

Figure 1) and this was the case (**Figure 2**). Although AMPAergic scaling was initially thought to play the role of spiking homeostat, it appears more likely that GABAergic scaling is one of the homeostatic mechanisms that is playing this role.

The results of our current study on GABAergic scaling and our previous study on AMPAergic scaling¹¹ suggest these two forms of plasticity have distinct triggers and signaling pathways. Optogenetic restoration of activity in CNQX prevented GABAergic downscaling (**Figure 3-4**) but had no effect on AMPAergic scaling¹¹. Further, increasing glutamatergic receptor activation with CTX during activity blockade reduced TTX-induced AMPAergic scaling¹¹ but not GABAergic scaling (**Figure 5D**). We considered the possibility that some of our results could be due to differences in the cultures of this vs our previous study (mouse vs rat, 20 vs 40 μ M CNQX, etc.). However, when we reduced spiking activity with 20 μ M CNQX and assessed AMPAergic scaling in mouse cortical cultures we did not trigger AMPAergic scaling at all, again consistent with the idea that the triggers are distinct for these two classes of plasticity. It is not clear to us why we were unable to trigger AMPAergic scaling in this study. It is possible that our cortical cultures (mouse, density) have less capacity for AMPAergic scaling. Alternatively, AMPAergic scaling may require higher concentrations of CNQX to partially influence NMDARs; this could occur through more complete blockade of AMPARs whose depolarization is important in removing the Mg block of the NMDAR or through direct block of the glycine binding site of the NMDAR^{30, 31}. Regardless, the reduction of spiking activity produced by 20 μ M CNQX was capable of triggering GABAergic scaling.

Previously, in embryonic motoneurons we found that both GABAergic and AMPAergic scaling was mediated by changes in GABAR activation from spontaneous release rather than changes in spiking activity^{12, 32}. However, this was at a developmental stage when GABA was depolarizing and could potentially activate calcium signaling pathways. On the other hand, spike rate homeostasis through the GABAergic system is consistent with many previous studies in which sensory input deprivation *in vivo* led to rapid compensatory disinhibition^{33, 34}. For instance, one day of visual deprivation (lid suture) reduced evoked spiking in fast spiking parvalbumin (PV) interneurons and this was thought to underlie the recovery of pyramidal cell responses to visual input at this point²⁵. One day of whisker deprivation between P17 and P20 produced a reduction of PV interneuron firing that was due to reduced intrinsic excitability in the GABAergic PV neuron²¹. In addition, one day after enucleation, the excitatory to inhibitory synaptic input ratio in pyramidal cells was dramatically increased due to large reductions in GABAergic inputs to the cell²⁴. This disinhibition occurs rapidly²³ and can outlast changes in glutamatergic counterparts^{22, 24}. These results highlight the important role that inhibitory interneurons play in the homeostatic maintenance of spiking activity. Further, these cells have extensive connectivity with pyramidal cells, placing them in a strong position to influence network excitability^{35, 36}. In the current study, we show a critical feature of homeostatic regulation of spiking is through one aspect of inhibitory control, GABAergic synaptic scaling in excitatory neurons.

It is not clear what specific features of spiking triggers GABAergic scaling. GABAergic scaling may require the reduction of spiking in multiple cells in a network, rather than a single cell. Reduced spiking with sporadic expression of a potassium channel in one hippocampal cell in culture did not induce GABAergic scaling in that cell¹⁷. Such a result could be mediated by the release of some activity-dependent factor from a collection of neurons. BDNF is known to be released in an activity-dependent manner and has been shown to mediate GABAergic downward scaling following activity block previously in both hippocampal and cortical cultures^{16, 37}. On the other hand, another study increased spiking in hippocampal cultures and showed that homeostatic increases in mIPSC amplitudes could be dependent on the individual cell's spiking activity¹⁸. Future work will be necessary to determine the exact feature of spiking that may be more critical in triggering GABAergic scaling (e.g. bursting vs total spike frequency) and the downstream signaling pathway (e.g. somatic calcium transients).

While we have no direct support of a role for NMDARs, we cannot rule out the possibility that NMDAR activity could contribute to GABAergic scaling. Previous work has shown that NMDAR block can trigger GABAergic downscaling ¹⁶ and our activity manipulations would similarly alter NMDAR activation (CNQX would reduce and optogenetic restoration would restore some NMDAR activation). Whatever the specific features of spiking activity that trigger GABAergic scaling, our results strongly point to the idea that GABAergic scaling, could serve a critical role of a spiking homeostat, and highlights the fundamentally important homeostatic nature of GABAergic neurons.

Finally, it is important to take into consideration some of the benefits and limitations of this study. By recording activity levels of cultured neurons through multi-electrode arrays we were able to identify the actual influence of the drugs on population activity. This is a step beyond what many homeostatic studies, including our own, typically do, and it affords us the opportunity to interpret more intelligently the results of our perturbations. Regardless, there are limitations associated with these techniques. Cultured networks lack the actual circuitry of the *in vivo* cortex, and for several reasons are vulnerable to dramatic variability (based on plating density, ages, composition, etc.).

This variability can be seen in response to drug application throughout our results and it is important to keep in mind that the recorded spiking activity represents the population response from many different classes of excitatory and inhibitory neurons, although the majority are thought to represent excitatory principal neurons. Despite these caveats, the culture system has allowed us to manipulate spiking activity in important ways, which has provided us the insight that GABAergic scaling is one of the homeostatic mechanisms that fulfills the expectations of a spike rate homeostat.

Materials and Methods

Cell Culture

Brain cortices were obtained from C57BL/6J embryonic day 18 mice from BrainBits or harvested from late embryonic cortices. Neurons were obtained after cortical tissue was enzymatically dissociated with papain. Cell suspension was diluted to 2,500 live cells per ml and 35,000 cells were plated on glass coverslips or planar MEA coated with polylysine (Sigma, P-3143) and laminin. The cultures were maintained in Neurobasal medium supplemented with 2% B27 and 2mM GlutaMax. No antibiotics or antimycotics were used. Medium was changed completely after one day *in vitro* (1 DIV) and half of the volume was then changed every 7 days. Spiking activity was monitored starting ~10 DIV to determine if a bursting phenotype was expressed and continuous recordings were made between 14-20 DIV. Cultures were discarded after 20 DIV. All protocols followed the National Research Council's Guide on regulations for the Care and Use of Laboratory Animals and from the Animal Use and Care Committee from Emory University.

Whole cell recordings

Pyramidal-like cells were targeted based on their large size. Whole-cell voltage clamp recordings of GABA mPSCs were obtained using an AxoPatch 200B amplifier, controlled by pClamp 10.1 software, low pass filtered at 5 KHz on-line and digitized at 20 KHz. Tight seals (>2 GΩ) were obtained using thin-walled boro-silicate glass microelectrodes pulled to obtain resistances between 7 and 10 MΩ. The intracellular patch solution contained the following (in mM): CsCl 120, NaCl 5, HEPES 10, MgSO₄ 2, CaCl₂ 0.1, EGTA 0.5, ATP 3 and GTP 1.5. The pH was adjusted to 7.4 with KOH.

Osmolarity of patch solution was between 280-300 mOsm. Artificial Cerebral-Spinal Fluid (ACSF) recording solution contained the following (in mM): NaCl 126, KCl 3, NaH₂PO₄ 1, CaCl₂ 2, MgCl₂ 1, HEPES 10 and D-glucose 25. The pH was adjusted to 7.4 with NaOH. GABAergic mPSCs were isolated by adding to ACSF (in μ M): TTX 1, CNQX 20 and APV 50. Membrane potential was held at -70 mV and recordings were performed at room temperature. Series resistance during recordings varied from 15 to 20 M Ω and were not compensated. Recordings were terminated whenever significant increases in series resistance (> 20%) occurred. Analysis of GABA mPSCs was performed blind to condition with MiniAnalysis software (Synaptosoft) using a threshold of 5 pA for mPSC amplitude (50 mPSCs were taken from each cell and their amplitudes were averaged and each dot in the scatterplots represent the average of a single cell). Ratio plots of mIPSCs were constructed by taking a constant total number of mIPSCs from control and drug-treated cultures (e.g. 15 control cells with 40 mIPSCs from each cell and 20 CNQX-treated cells with 30 mIPSCs from each cell, 600 mIPSCs per condition). Then the amplitudes of mIPSCs from each condition were rank ordered from smallest to largest and plotted as a ratio of the drug-treated amplitude divided by the control amplitude, as we have described previously [4](#), [26](#), [38](#).

MEA recordings

Extracellular spiking was recorded from cultures plated on planar 64 channel MEAs (Multichannel Systems) recorded between 14-20 DIV in Neurobasal media with B27 and GlutaMax, as described above. Cultured MEAs were covered with custom made MEA rings with gas permeable ethylene-propylene membranes (ALA Scientific Instruments). Synapse software (Tucker-Davis Technologies TDT) was used to monitor activity on a TDT electrophysiological platform consisting of the MEA MZ60 headstage, the PZ2 pre-amplifier and a RZ2 BioAmp Processor. Recordings were band-pass filtered between 200 and 3000Hz and acquired at 25KHz. MEA's were placed in the MZ60 headstage, which was housed in a 5% CO₂ incubator at 37°C. Drugs were added separately in a sterile hood and then returned to the MEA recording system. MEA spiking activity was analyzed offline with a custom-made Matlab program. The recordings acquired in Synapse software (TDT) were subsequently converted using the subroutine TDT2MAT (TDT) to Matlab files (Mathworks). The custom written Matlab program identified bursts of network spikes using an interspike interval-threshold detection algorithm [39](#). Spiking activity was labeled as a network burst when it met a user-defined minimum number of spikes (typically 10) occurring across a user-defined minimum number of channels (5-10) within a Time-Window (typically 0.1-0.3 seconds) selected based on the distribution of interspike intervals (typically between the first and 10th consecutive spike throughout the recording, Supplemental Figure 2). This program allowed us to remove silent channels and channels that exhibited high-noise levels. The identified network bursts were then visually inspected to ensure that these parameters accurately identified bursts. The program also computed network burst metrics including burst frequency, overall spike frequency and other characteristics.

Optogenetic control of spiking

For photostimulation experiments neurons were plated on 64-channel planar MEAs and transfected with AAV9-hSynapsin-ChR2(H134R)-eYFP (ChR2) produced by the Emory University Viral Vector Core. All cultures used in ChR2 experiments, including controls, were transfected at 1 DIV. The genomic titer was 1.8×10^{13} vg/ml. Virus was diluted 1 to 10,000 in growth medium and this dilution was used for the first medium exchange at DIV 1. Finally, the media containing the virus was washed out after 24 hour incubation. A 3 hour pre-drug recording was obtained in the TDT program that determined the average MEA-wide firing rate before adding CNQX. This custom written program from TDT then delivered a TTL pulse (50-100ms) that drove a blue light photodiode (465 nm, with a range from 0 to 29.4 mwatts/mm², driven by a voltage command of 0-4V) from a custom-made control box that allowed for scaled illumination. The photodiode sat directly below the MEA for activation of the ChR2. This triggered a barrage of spikes resulting in a burst that looked very similar to a naturally occurring burst not in the presence of CNQX. The

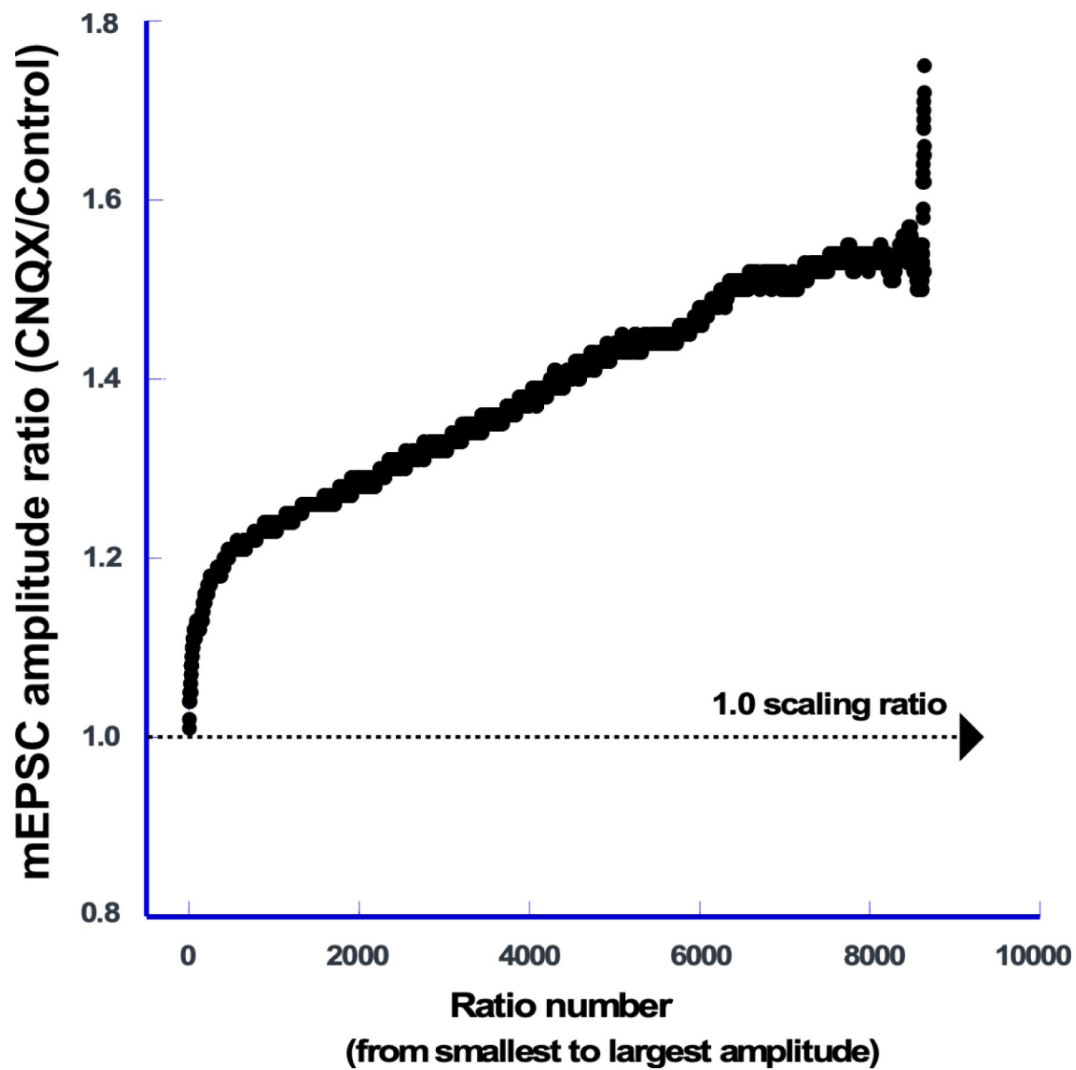
program measured the MEA-wide spike rate every 10 seconds and if the rate fell below the set value established from the pre-drug average, an optical stimulation (50-100ms) was delivered triggering a burst which then increased the average firing rate, typically above the set point.

Statistics

Estimation statistics have been used throughout the manuscript. 5000 bootstrap samples were taken; the confidence interval is bias-corrected and accelerated. The P value(s) reported are the likelihood(s) of observing the effect size(s), if the null hypothesis of zero difference is true. For each permutation P value, 5000 reshuffles of the control and test labels were performed (Moving beyond P values: data analysis with estimation graphics ⁴⁰[↗](#)).

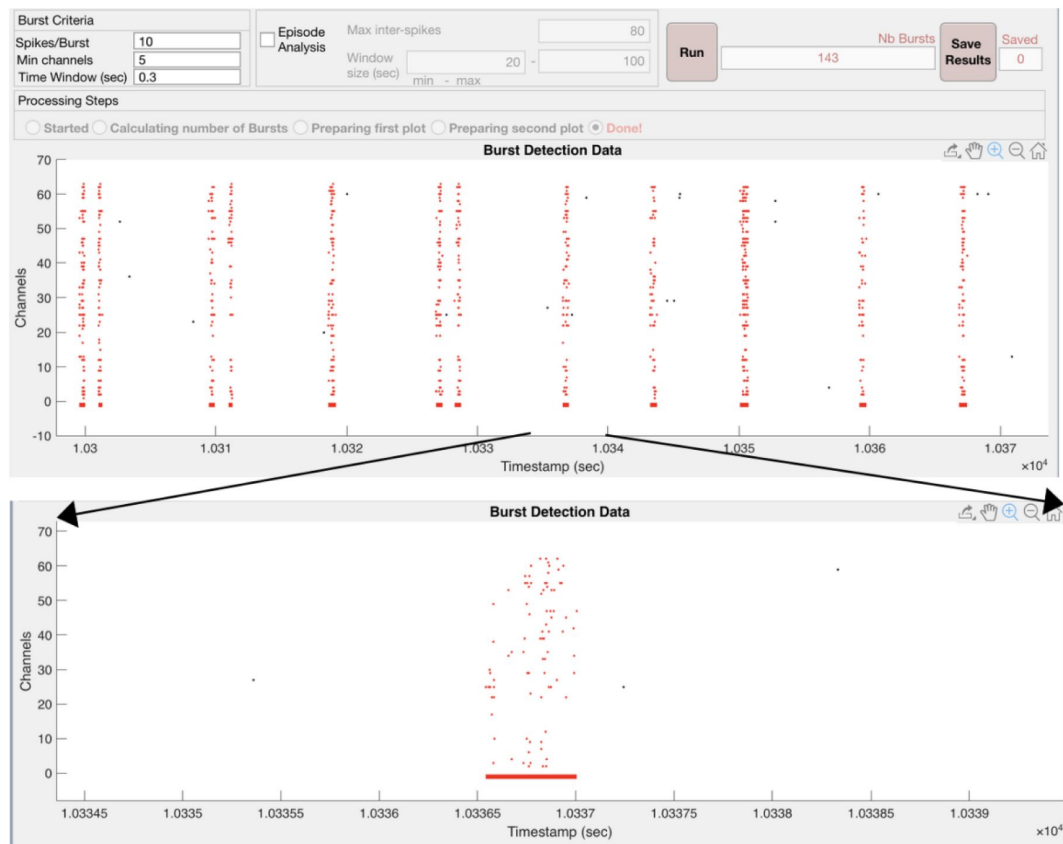
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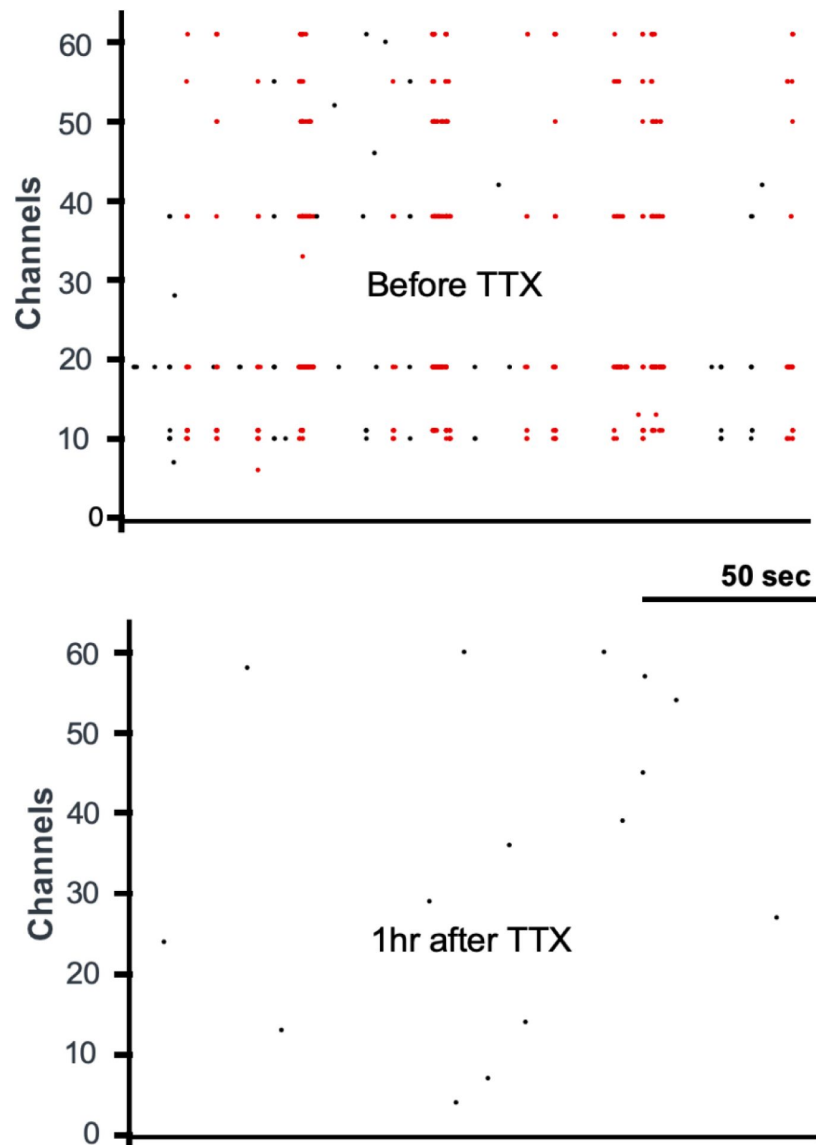
Supplemental Figure 1.

AMPA block triggered non-uniform AMPAergic scaling. Scaling ratio plot shows the ratio of rank ordered mEPSC amplitudes from CNQX-treated cultures (n=95 cells, 91 mEPSCs/cell) divided by those from untreated cultures (n=91 cells, 95 mEPSCs/cell). The X axis represents the rank ordered number of mEPSCs (from smallest to largest).



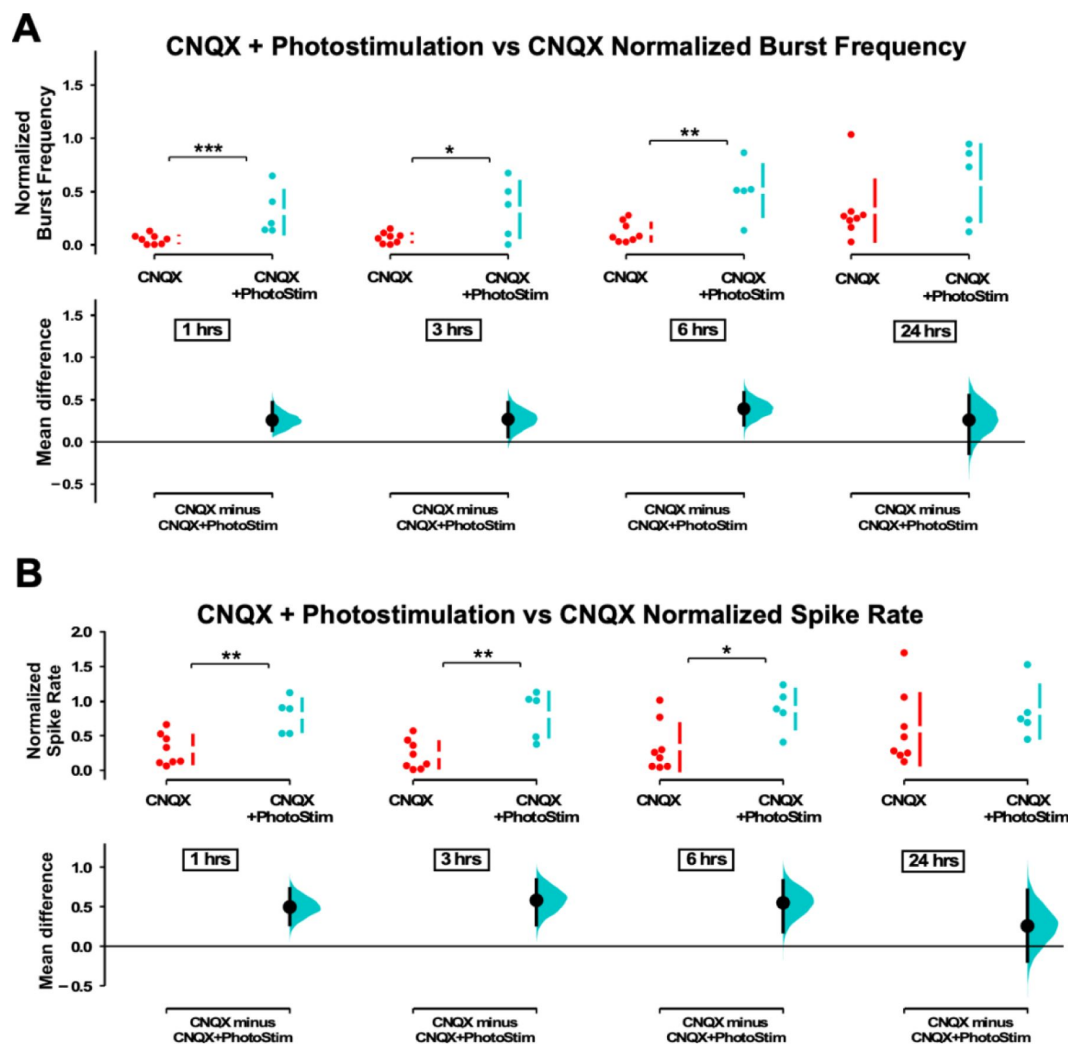
Supplemental Figure 2.

Custom written Matlab program identifies bursts in cortical cultures plated on MEA's by choosing the minimum number of spikes per burst (Spikes/Burst) across a minimum number of channels contributing to a burst (Min channels) within a maximum Time Window. Upper image shows the identification of bursts in red across 64 channels as a raster plot where each dot represents one spike detected on the MEA. The program then examines various parameters which were then exported to an excel spreadsheet for analysis. Burst identity and duration are shown as a red line positioned below the raster plot. A single burst is expanded and plotted below the upper image.



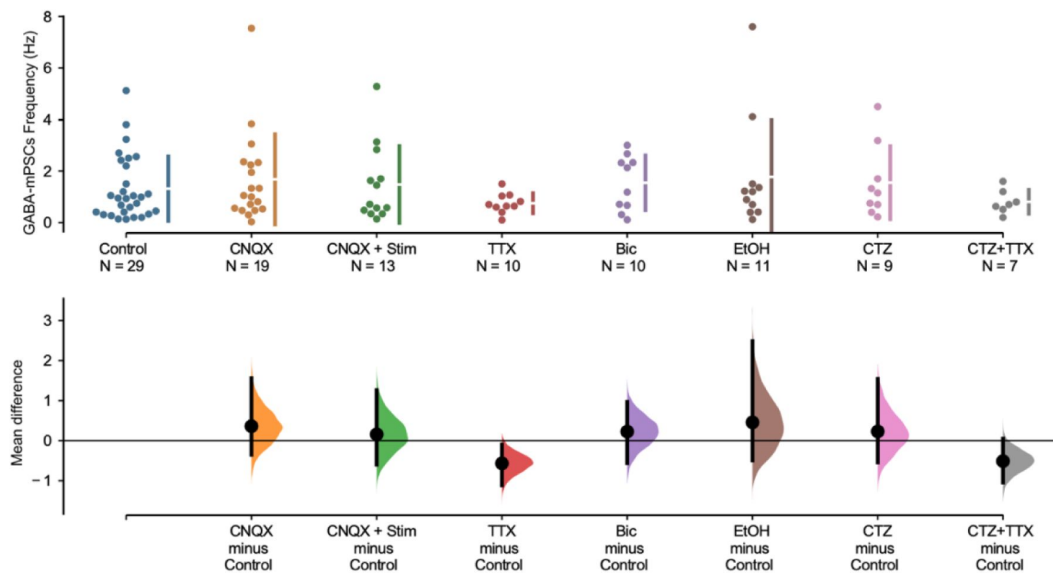
Supplemental Figure 3.

Raster plot of cortical culture plated on MEA demonstrating network bursting (red dots, upper plot). Bursts were then abolished after addition of TTX ($1\mu\text{M}$) to the culture; a small number of spike detections remain, however these are likely to be noise that crosses the detection threshold.



Supplemental Figure 4.

MEA recording's show optostim + CNQX increases burst frequency and spike frequency compared to CNQX alone. **A)** Average burst rate is compared for CNQX-treated cultures with optogenetic stimulation ($n=5$) and CNQX only unstimulated cultures ($n=8$) at 1hr, 3hrs, 6hrs, and 24hrs ($p=0.209$) after addition of CNQX. **B)** Average overall spike frequency is compared for CNQX-treated cultures with optogenetic stimulation and CNQX only unstimulated cultures at 1hr, 3hrs, 6hrs, and 24hrs ($p=0.389$) after addition of CNQX. The mean differences at different time points are compared to control and displayed in Cumming estimation plots. Significant differences denoted by * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$. Recordings from single cultures (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. Mean differences between control and treated groups are plotted on the bottom panel, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars).



Supplemental Figure 5.

Frequency of mIPSCs were no different across conditions. Scatterplots of mIPSC frequency show tremendous variability but do not exhibit significant differences through different drug treatments. The mean differences are compared to control and displayed in Cumming estimation plots. GABAergic mIPSC frequencies from single neurons (filled circles), where mean values (represented by the gap in the vertical bar) and SD (vertical bars) are plotted on the upper panels. Mean differences between control and treated groups are plotted on the bottom panel, as a bootstrap sampling distribution (mean difference is represented by a filled circles and the 95% CIs are depicted by vertical error bars).

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Reviewer #3 (Public Review):

This paper concerns whether synaptic scaling (or homeostatic synaptic plasticity; HSP) occurs similarly at GABA and Glu synapses and comes to the surprising conclusion that these can be regulated independently. In fact, under the conditions used in this study, only the GABAergic synapses show HSP and the glutamatergic synapses don't change. This is surprising because these were thought to be co-regulated during HSP and in fact, the major mechanisms thought to underlie downscaling (TTX or CNQX driven), retinoic acid and TNF, have been shown to regulate both GABARs and AMPARs directly. Thus, the main result, that GABA HSP is dissociable from Glu HSP, is novel and exciting. This suggests either different mechanisms underlie the two processes, or that under certain conditions, another mechanism is engaged that scales one type of synapse and not the other. Given that glutamatergic synapses are unchanged in their conditions, that later seems more likely - a novel form of HSP exists that only scale GABAergic synapses. Whether glutamatergic and GABAergic synapses scale independently during HSP affecting both types of synapses remains to be addressed. It would be necessary to demonstrate the dissociation in the same system, under conditions where both types of synapses are changing. But because the form of HSP studied here appears different than that studied in Fong et al., the authors should be careful when comparing the two results. There seems to be an implicit underlying assumption that there is a simple form of HSP, when the overall literature (and the two studies from this lab) supports the idea of many forms of HSP.

The homeostatic changes at GABAergic synapses do seem to be more consistent in amplitude across the bulk of the synapses, which does suggest that true scaling (a proportional change to all synapses on a cell) is occurring. This may represent a major difference in how homeostatic changes occur at the two types of synapses.

The second finding is that this form of HSP seems more regulated by action potential firing than conventional HSP - previous work from this lab had shown that restoring AP firing during AMPA receptor blockade did not prevent scaling of glutamatergic synapses (it should be noted these experiments were done in rat cultures, not mouse, used a higher concentration of CNQX, and used a different optogenetic stimulation paradigm). Restoring AP firing rates under the conditions used here (and thus the form of HSP only affecting GABA synapses), on the other hand, did prevent the homeostatic response. This suggests that this GABA-only form of HSP is more attuned to spiking rates than other forms.

However, details in the data may suggest that spiking is not the (or the only) homeostat, as TTX and CNQX causes identical changes in mIPSC amplitude but have different effects on spiking (although TTX may be driving a different form of HSP). Further, in Fig 5, CTZ had a minimal effect on spiking but a large effect on mIPSCs. Similar issues appear in Fig 6, where the induction of increased spiking is highly variable, with many cells showing control levels or lower spiking rates. Yet the synaptic changes are robust, across all cells. Overall, more will need to be done to conclude that spiking is the homeostat for GABA synapses.

The paper also suggests that the GABA changes are leading to the recovery of the spiking rates, but while they have the time course of the spiking changes and recovery, they only have the 24h time point for synaptic changes. It is not yet possible to conclude how the time courses align without more data, nor can we assume that cells that did not recover to control firing rates would do so eventually.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment

This study assesses homeostatic plasticity mechanisms driven by inhibitory GABAergic synapses in cultured cortical neurons. The authors report that up- or down-regulation of GABAergic synaptic strength, rather than excitatory glutamatergic synaptic strength, is critical for homeostatic regulation of neuronal firing rates. The reviewers noted that the findings are potentially important, but they also raised questions. In particular, the evidence supporting the findings is currently incomplete and demonstration of independent regulation of mEPSCs and mIPSCs is a necessary experiment to support the major claims of the study.

We appreciate the detailed, thoughtful assessment of our paper by the reviewers and editors and now submit a revised version that addresses the reviewers' comments as detailed below in response to each concern. We include a more open discussion of alternative possibilities and have added experiments demonstrating that AMPAergic scaling in our mouse cortical cultures is triggered differently than GABAergic scaling. We treated the cultured neurons exactly as described for triggering GABAergic scaling (20 μ M CNQX for 24 hours), however this did not trigger AMPAergic upscaling (new Figure 7), even though it did reduce spiking/bursting activity. Below we explain the result further, but ultimately this does demonstrate independent regulation of mEPSCs and mIPSCs as requested by the editor/reviewer (spike reductions induced by CNQX reduced mIPSC amplitude, but had no effect on mEPSC amplitude).

Reviewer #1 (Public Review):

While the paper is ambitious in its rhetorical scope and certainly presents intriguing findings, there are several serious concerns that need to be addressed to substantiate the interpretations of the data. For example, the CTZ data do not support the interpretations and conclusions drawn by the authors. Summarily, the authors argue that GABAergic scaling is measuring spiking (at the time scale of the homeostatic response, which they suggest is a key feature of a homeostat) yet their data in figure 5B show more convincingly that CTZ does not influence spiking levels - only one out of four time points is marginally significant (also, I suspect that the bootstrapping method mentioned in line 454-459 was conducted as a pairwise comparison of distributions. There is no mention of multiple comparisons corrections, and I have to assume that the significance at 3h would disappear with correction).

We certainly understand the criticism here (similar to reviewer 2's third point). We now discuss these complications in a more detailed description in the manuscript (CTZ section of results and at end of the discussion). First, we are presenting our entire dataset to be as transparent as possible. Unlike most synaptic scaling studies (including our own) that apply drugs to alter activity and assess mPSC amplitude at the final time point, here we are actually showing CTZ's effect on spiking activity within the culture over time. This is critical because it has informed us of the drug's true effect on spiking, the variability that is associated with these perturbations, and the ability and timing of the cultured network to homeostatically recover initial levels. This was important because it revealed that the drugs do not always influence activity in the way we assume, and this provides greater context to our results. Second, we are showing all of our data, and presenting it using estimation statistics which go beyond the dichotomy of a simple p value yes or no (Ho J, Tumkaya T, Aryal S, Choi H, Claridge-Chang A. 2019. Moving beyond P values: data analysis with estimation graphics. Nat

Methods 16: 565-66). Estimation statistics have become a more standard statistical approach in the last 15 years and is the preferred method for the Society for Neuroscience's eNeuro Journal. This method shows the effect size and the confidence interval of the distribution. For the 3 hr time point in Fig. 5B the CTZ/ethanol vs. ethanol data points exhibit very little overlap and the effect size demonstrates a near doubling of spike frequency, and the confidence interval shows a clear separation from 0. This was a pairwise comparison as we compared values at each time point after the addition of ethanol or ethanol/CTZ. Third, the plots illustrate an upward trend in spike frequency at 1 and 6 hrs, but that there is also clear variability. It is important to note that these are multiunit recordings and not purely excitatory principal neurons that we target for mPSC recordings. This complication along with the variability inherent in these cultures could make simple comparisons difficult to interpret and we now discuss this (end of discussion). Regardless, we do see some increase in spiking with CTZ and we clearly see increases in mIPSC amplitude, thus providing some support for the idea that spiking could be a critical player in terms of GABAergic scaling, particularly when put in the context of all of our findings. Future work will be necessary to determine how alterations in spiking lead to changes in mIPSC amplitude and we now discuss this (2nd to last paragraph in discussion).

Then, the fact that TTX applied on top of CTZ drives an increase in mIPSC amplitude is interpreted as a conclusive demonstration that GABAergic scaling is sensing spiking. It is inevitable, however, that TTX will also severely reduce AMPAR-R activation - a very plausible alternative explanation is that the augmentation of AMPAR activation caused by CTZ is not sufficient to overcome the dramatic impact of TTX. All together, these data do not provide substantial evidence for the conclusion drawn by the authors.

We believe that the most parsimonious explanation for our results is that spiking activity, not AMPAR activation, triggers GABAergic downscaling. GABAergic scaling is no different when comparing 24hr TTX treatment vs TTX+CTZ, and optogenetic restoration of spiking activity while continuing to block AMPAR activation was able to restore GABAergic mPSC amplitudes to control levels. It is important to emphasize that our results with TTX vs. TTX+CTZ are different for GABAergic scaling (no difference in this study) and AMPAergic scaling (CTZ diminished upward scaling in previous study – Fong et al., 2015 - PMID: 25751516) suggesting different triggers for the two forms of scaling. While we strongly believe we have demonstrated that GABAergic downscaling is dependent on spiking (not AMPAergic transmission), we now acknowledge that we cannot rule out the possibility that upward GABAergic scaling may be influenced by AMPAR activation (2nd paragraph discussion), although we have no evidence in support of this.

Specific points:

- The logic of the basis for the argument is somewhat flawed: A homeostat does not require a multiplicative mechanism, nor does it even need to be synaptic. Membrane excitability is a locus of homeostatic regulation of firing, for example. In addition, synapse-specific modulation can also be homeostatic. The only requirement of the homeostat is that its deployment subserves the stabilization of a biological parameter (e.g., firing rate).

We largely agree with the reviewer and should not have implied that this was a necessary requirement for a spike rate homeostat. What we should have said was that historically this definition has been applied to AMPAergic scaling, which is thought to be a spike rate homeostat. We have now corrected this (introduction and discussion).

- Line 63 parenthetically references an important, but contradictory study as a brief "however". Given the tone of the writing, it would be more balanced to give this study at

least a full sentence of exposition.

Agreed, and we have now done this.

- The authors state (line 11) that expression of a hyperpolarizing conductance did not trigger scaling. More recent work ('Homeostatic synaptic scaling establishes the specificity of an associative memory') does this via expression of DREADDs and finds robust scaling.

The purpose of citing this study was to argue that the spike rate homeostat hypothesis doesn't make sense for AMPAergic scaling based on a study that hyperpolarized an individual cell while leaving the rest of the network unaltered and therefore leaving network activity and neurotransmission largely normal. In this previous study scaling was not triggered, suggesting reduced spike rate within an individual cell was insufficient to trigger scaling in that cell. The more recent study mentioned by the reviewer achieved scaling by hyperpolarizing a majority of cells in the network. Importantly, this approach alters neurotransmission throughout the network, making it challenging to isolate the specific contributions of spiking vs. receptor activation. Unlike the previous study, which focused on the impact within individual cells, this newer study involves global alterations in network activity, complicating the interpretation of the role of spiking versus receptor activation in triggering scaling.

- Supplemental figure 1 looks largely linear to me? Out of curiosity, wouldn't you expect the left end to be aberrant because scaling up should theoretically increase the strength of some synapses that would have been previously below threshold for detection?

We agree that the scaling ratio plot is largely linear. To be clear, the linearity of the ratio plot was not our point here, rather that there was a positive slope meaning ratios (CNQX mEPSC amplitudes/control mEPSC amplitudes) got bigger for the larger CNQX-treated mEPSCs. Alternatively, a multiplicative relationship where mEPSCs are all increased by a single factor (e.g. 2X) would be a flat line with 0 slope at the multiplicative value (e.g. 2). In terms of the left side of the plot, we do see values that rise abruptly from 1 - this was partially obstructed by the Y axis in this figure and we have adjusted this. This left part of the plot is likely due the CNQX-induced increases in mEPSC amplitudes of mini's that were below our detection threshold of 5pA, as suggested by the reviewer. Therefore, mini's that were 4pAs could now be 5pAs after CNQX treatment and these are then divided by the smallest control mEPSCs which are 5 pAs (ratio of 1). We tried to do a better job describing this in the resubmission (1st paragraph of results).

- Given that figure 2B also shows warping at the tail ends of similar distributions, how is this to be interpreted?

The left side of the ratio plot shows evidence consistent with the idea that mIPSCs are dropping into the noise after CNQX treatment (smallest GABA mIPSCs that don't fall into noise are 5pA and this is divided by the smallest control GABA mPSCs of 5pPA and therefore the ratio is 1). The rest of the distribution will then approach the scaling factor (50% in this case). On the right side of the ratio plot the values appear to slightly increase. We are not sure why this is happening, but it maybe that a small percentage of mIPSCs are not purely multiplicative at 0.5, however the biggest mPSCs can vary to a great degree from one cell to the next and in other cases we do not see this (Figure 4B, Figure 5E). We tried to do a better job describing this in the resubmission (results describing Figure 2).

- The readability of the figures is poor. Some of them have inconsistent boundary boxes, bizarre axes, text that appears skewed as if the figures were quickly thrown together and

stretched to fit.

We have adjusted the figures to be more consistent throughout the manuscript.

- I'm concerned about the optogenetic restoration of activity experiment. Cortical pyramidal neuron mean firing rates are log normally distributed and span multiple orders of magnitude. The stimulation experiments can only address the total firing at a network-level - given than a network level "mean" is meaningless in a lognormal distribution, how are we to think about the effect of this manipulation when it comes to individual neurons homeostatically stabilizing their own activities? In essence, the argument is made at the single-neuron level, but the experiment is conducted with a network-level resolution.

As described above, we do not have the capacity to know what the actual firing rate of a particular neuron was before and after perturbing the system, and certainly not for the specific cells we recorded from to obtain mPSC amplitudes, and so we cannot say that we have perfectly restored the original firing rates of neurons. However, there is reason to believe that this is achieved to some extent. Our optogenetic stimulation is only 50-100 ms long activating a subset of neurons. This is sufficient to provide a synaptic barrage that then triggers a full blown network burst where the majority of spikes occur, but this is after the light is off. In other words, the optogenetic light pulse only initiates what becomes a relatively normal network burst that fortunately allows the individual cells to express their relatively normal (pre-drug) activity pattern. In our previous study using optogenetic activity restoration (Fong et al., 2015) we were able to show that this was the case for individual units - the spiking of an individual unit during a burst is similar before and after CNQX/optogenetic stimulation (see Figure 4b and Suppl. Fig 4 in Fong et al. 2015). We are not claiming that we have restored spiking to exactly the pre-drug state, but bring it back toward those levels and we see this is associated with a return of the mIPSC amplitude to near control levels. We now include a brief description of this in the manuscript (results describing Figure 3).

- Line 198-99: multiplicativity is not a requirement of a homeostatic mechanism.

- Line 264-265 - again, neither multiplicativity and synaptic mechanisms are fundamentally any more necessary for a homeostatic locus than anything else that can modulate firing rate in via negative feedback.

As mentioned above, the multiplicative nature of scaling has been a historical proposal for AMPAergic scaling and we have now found such a relationship for GABAergic scaling. This is important for understanding how this plasticity works, but we agree that it is not necessary for a homeostat and we have adjusted the manuscript accordingly.

- 277: do you mean AMPAR?

We were not clear enough here. We actually do mean GABAR. The idea was that CTZ increases network activity and thus increases both AMPAergic and GABAergic transmission. We have rewritten this part of the discussion to avoid any confusion (2nd paragraph discussion).

- Example: Figure 1A is frustratingly unreadable. The axes on the raster insets are microscopic, the arrows are strangely large, and it seems unnecessary to fill so much realstate with 4 rasters. Only one is necessary to show the concept of a network burst. The effect of time+CNQX on the frequency of burst is shown in B and C.

- Example: Figure 2 appears warped and hastily assembled. Statistical indications are shown within and outside of bounding boxes. Axes are not aligned. Labels are not

aligned. Font sizes are not equal on equivalent axes.

These figures were generated by the estimation statistics website and text may have been resized inappropriately. We have tried to adjust this and now have attempted to standardize the axes text to the best of our ability.

- The discussion should include mention of the limitations and/or constraints of drawing general conclusions from cell culture.

We have added this consideration at the end of the discussion. Further, this is why we cited studies that argue GABAergic neurons have a particularly important role in homeostatic regulation of firing following sensory deprivations *in vivo*.

- The discussion should include mention of the role of developmental age in the expression of specific mechanisms. It is highly likely that what is studied at ~P14 is specific to early postnatal development.

We now discuss caveats of cortical cultures at the end of the discussion.

It is essential to ensure that the data presented in the paper adequately supports the conclusions drawn. A more cautious approach in interpreting the results may lead to a stronger argument and a more robust understanding of the underlying mechanisms at play.

We have broadened our discussion of alternative interpretations throughout the manuscript.

Reviewer #1 (Recommendations For The Authors):

*While I am hesitant to judge a paper based on its tone, I would personally recommend revision of some of the subjective words and statements, as the manuscript undermines its own effectiveness by making unnecessarily strong statements. The text repeatedly paints an "either A or B" picture, and if there's any general lesson in biology, it's that it's always A and B. Global, multiplicative glutamatergic scaling could quite conceivably occur alongside GABAergic scaling, as well as synapse-specific homeostatic modifications. It seems that it would be wise to acknowledge that, while the data presented here point in one direction, *in vivo* results in an adult brain (for example) might present an entirely different set of patterns. This will not only enhance the readability of the paper but also ensure that the scientific community can engage with the work in a constructive and collaborative manner. Again, I present this as only a constructive and supportive suggestion. I am a big fan of work from this laboratory, and I would love to see this paper in an improved form - it's an important set of ideas and I do believe that these data are rigorously collected.*

We have attempted to provide a more comprehensive interpretation of our results. We agree that a homeostat can come in many flavors, but do believe that GABAergic scaling is strong candidate, whereas AMPAergic scaling does not currently fit such a role. We do now discuss caveats with our work and are open to other interpretations that need to be flushed out in future work.

Reviewer #2 (Public Review):

Major points:

(1) The reason why CNQX does not completely eliminate spiking is unclear (Fig. 1). What is the circuit mechanism by which spiking continues, although at lower frequency, in the

absence of AMPA-mediated transmission and what the mechanism by which spiking frequency grows back after 24h (still in the absence of AMPA transmission)?

Is it possible that NMDA-mediated transmission takes over and triggers a different type of network plasticity?

The bursting in AMPAR blockade is due to the remaining NMDA receptor-mediated transmission. We showed this in our previous study in Suppl. Figure 2 and 6 of Fong et al., 2015 (PMID: 25751516). Our ability to optically induce normal looking bursts of spikes was also dependent NMDAR activation (Fong et al 2015 and Figure 6 Newman et al., 2015 - PMID: 26140329). Further, in Dr Fong's PhD dissertation it was shown that the bursting activity was abolished when AMPA and NMDA receptors were both blocked. There are likely many factors that contribute to the recovery of activity, and certainly one of them is likely to be the weakening of inhibitory GABAergic currents as we had mentioned. We have now added the point about NMDARs mediating the remaining bursts in the manuscript (results associated with Figure 1). We are not clear on what the reviewer has in mind in terms of "NMDA-mediated transmission takes over and triggers a different kind of network plasticity", but we do discuss the possibility that spiking triggers GABAergic scaling through its effect on NMDAergic transmission, which we cannot rule out, but also have no evidence in support of this idea (3rd and 5th paragraph of discussion). We do plan on addressing this in a future work.

(2) A possible activation of NMDARs should be considered. One would think that experiments involving chronic glutamatergic blockade could have been conducted in the presence of NMDAR blockers. Why this was not the case?

Unfortunately, it was not possible to optogenetically restore normal bursting in the presence of NMDAR blockade (even when AMPAergic transmission was intact), as NMDARs appeared to be critical for the optical restoration of the normal duration and form of the burst in rat cortical cultures (see Suppl. Figure 6 Fong et al., 2015 Nat Comm and Figure 6 Newman et al., 2015). Even high concentrations of CNQX (40 μ M) prevented us from restoring spiking in mouse cultures in the current study, which is why we moved to 20 μ M CNQX for this study. The reviewer raises an excellent point about a possible NMDAR contribution to altered synaptic strength, however. It is likely that NMDAR signaling is reduced in the presence of CNQX since burst frequency was dramatically reduced along with AMPAR-mediated depolarizations. We cannot rule out the possibility that NMDAR signaling could contribute to the alterations in GABAergic mIPSCs and discuss this in the resubmission (3rd and 5th paragraph of the discussion). We had not considered this previously because prior work suggested that 24/48 hour block NMDARs (APV) did not trigger AMPAergic scaling in cortical or hippocampal cultures (see Figure 1 Turrigiano et al., 1998 Nature and Suppl. Figure 4 Sutton et al., 2006 Cell), moreover, our previous study showed that restoring NMDAergic transmission ontogenetically, at least to some extent, had no influence on AMPAergic scaling (Fong et al., 2015).

Also, experiments with global Chr2 stimulation with coincident pre and postsynaptic firing might also activate NMDARs and result in additional effects that should be taken into consideration for the global scaling mechanism.

To be clear, our optical stimulation was of short duration (duration 50-100 ms) and was turned off before the vast majority of spiking that occurred in the bursts. So the light flash was a trigger that allowed a relatively normal looking burst to occur after the light was off (see lower panel of Figure 3B optogenetic stimulation – short duration only at onset of burst – we now make this clearer in resubmission). Therefore, we were unlikely to trigger significant synchronous activation that does not normally occur in network bursts.

(3) Cultures exposed to CTZ to enhance AMPA receptors generated variable results (Fig. 5), somewhat increasing spiking activity in a non-significant manner but, at the same time, strengthening mIPSC amplitude. This result seems to suggest that spiking might be involved in GABAergic scaling, but it does not seem to prove it. Then, addition of TTX that blocked spiking reduced mIPSC amplitude. It was concluded here that the ability of CTZ to enhance GABAergic currents was primarily due to spiking, rather than the increase in AMPA-mediated currents. However, in addition to blocking action potentials, TTX would also prevent activation of AMPARs in the presence of CTZ due to the lack of glutamatergic release. Therefore, under these conditions, an effect of glutamatergic activation on GABAergic scaling cannot be ruled out.

These concerns were very similar to reviewer 1's first comments (see above). To be clear we are going a step beyond most scaling studies by assessing MEA-wide firing rate, but this still provides an incomplete picture of the particular cells that we target for patch recordings in terms of their firing before and after a drug. Further, we see considerable variability in effect on firing rate from culture to culture, which we now discuss in the resubmission (final paragraph discussion). The fact that mIPSCs are no different after TTX treatment vs CTZ+TTX treatment suggests that AMPAergic transmission is not so influential on GABAergic downscaling. While the CTZ results are not conclusive by themselves, taken together with the optogenetic results, where restoration of spiking in AMPAR blockade reverses scaling, is most consistent with idea that GABAergic scaling is triggered by spiking rather than AMPAR activation and places GABAergic scaling as a strong candidate as spike rate homeostat. Although we do feel that we have demonstrated that downward GABAergic scaling is dependent on spiking, we cannot rule out the possibility that upward GABAergic scaling could be influenced by AMPAR activation to some extent. We now acknowledge this possibility (2nd paragraph discussion).

(4) The sample size is not mentioned in any figure. How many cells/culture dishes were used in each condition?

The individual dots represent either individual cells for mIPSC amplitude or individual cultures in MEA experiments. Number of cultures and cells are now stated in the figure legends.

(5) Cortical cultures may typically contain about 5-10% GABAergic interneurons and 90-95 % pyramidal cells. One would think that scaling mechanisms occurring in pyramidal cells and interneurons could be distinct, with different impact on the network. Although for whole-cell recordings the authors selected pyramidal looking cells, which might bias recordings towards excitatory neurons, naked eye selection of recording cells is quite difficult in primary cultures. Some of the variability in mIPSC amplitude values (Fig. 2A for example) might be attributed to the cell type? One could use cultures where interneurons are fluorescently labeled to obtain an accurate representation. The issue of the possible differential effects of scaling in pyramidal cells vs. interneurons and the consequences in the network should be discussed.

We now include this discussion in the resubmission (final paragraph discussion). Briefly, we chose large cells, which will be predominantly glutamatergic neurons as suggested by the reviewer. Ultimately, even among glutamatergic principal cells there may be variability in the response to drug application. All of these issues could contribute to variability and we have expanded our description of the variability in our results, including that based on cellular heterogeneity.

Reviewer #2 (Recommendations For The Authors):

Minor comments –

Fig S3: Please quantify changes in frequency

We have done this (Supplemental Figure 5).

Fig 2: please choose colors with higher contrast for CNQX/TTX

We have done this.

Fig. 3C: Why doesn't CNQX+PhotoStim reach control levels of bursting at 2h?

The program was designed to follow and maintain total spike frequency and so it does a better job at this than maintaining burst frequency.

Fig. 5A: please include a comparison between control and Ethanol

We now do this in Figure 5C. Both around 26pAs.

Fig. 5C: where is the Etoh condition?

We have made this figure more clear in terms of controls (Figure 5C & D).

Reviewer #3 (Public Review):

This paper concerns whether scaling (or homeostatic synaptic plasticity; HSP) occurs similarly at GABA and Glu synapses and comes to the surprising conclusion that these are regulated separately. This is surprising because these were thought to be co-regulated during HSP and in fact, the major mechanisms thought to underlie downscaling (TTX or CNQX driven), retinoic acid and TNF, have been shown to regulate both GABARs and AMPARs directly. (As a side note, it is unclear that the manipulations used in Joseph and Turrigiano represent HSP, and so might not be relevant). Thus the main result, that GABA HSP is dissociable from Glu HSP, is novel and exciting. This suggests either different mechanisms underlie the two processes, or that under certain conditions, another mechanism is engaged that scales one type of synapse and not the other.

However, strong claims require strong evidence, and the results presented here only address GABA HSP, relying on previous work from this lab on Glu HSP (Fong, et al., 2015). But the previous experiments were done in rat cultures, while these experiments are done in mice and at somewhat different ages (DIV). Even identical culture systems can drift over time (possibly due to changes in the components of B27 or other media and supplements). Therefore it is necessary to demonstrate in the same system the dissociation. To be convincing, they need to show the mEPSCs for Fig 4, clearly showing the dissociation. Doing the same for Fig 5 would be great, but I think Fig 4 is the key.

We understand the concern of the reviewer as we do see significant variability within our cultures and they were plated in different places, by different people, in different species (rat vs mouse). Therefore, we have attempted to redo the study on AMPAergic scaling on these mouse cortical neurons. Surprisingly, we found that 20 μ M CNQX did not trigger AMPAergic upscaling (new Figure 7), even though it did reduce spiking activity and was able to produce GABAergic downscaling. We did not carry out the optogenetic restoration of activity, because we did not trigger upscaling. The result does however, show that the reductions in

spiking/bursting that trigger GABAergic downscaling, did not trigger AMPAergic upscaling and therefore dissociate the 2 forms of scaling in these mouse cultures. We do not know why 20 μ M CNQX did not trigger scaling in these cultures since it does reduce spiking and AMPAR activation. In the Fong study we used 40 μ M CNQX because intracellular recordings from rat cortical neurons suggested this was required to completely block AMPAergic currents. Our initial studies in the current manuscript examining GABAergic scaling in mouse cortical cultures used 40 μ M CNQX, however, this concentration of CNQX prevented us from restoring spiking through optogenetic activation, so we reduced our concentration to 20 μ M CNQX, which did trigger GABAergic downscaling and allowed the restoration of spiking. We now show and discuss this result (Figure 7 and 3rd paragraph discussion).

The paper also suggests that only receptor function or spiking could control HSP, and therefore if it is not receptor function then it must be spiking. This seems like a false dichotomy; there are of course other options. Details in the data may suggest that spiking is not the (or the only) homeostat, as TTX and CNQX causes identical changes in mIPSC amplitude but have different effects on spiking. Further, in Fig 5, CTZ had a minimal effect on spiking but a large effect on mIPSCs. Similar issues appear in Fig 6, where the induction of increased spiking is highly variable, with many cells showing control levels or lower spiking rates. Yet the synaptic changes are robust, across all cells. Overall, this is not persuasive that spiking is necessarily the homeostat for GABA synapses.

Together our results argue against AMPAR or GABAR activation as a trigger for GABAergic scaling and that this is different than our results for AMPAergic scaling. These points alone are important to recognize. While changes in spiking do not perfectly follow the changes in GABAergic scaling they do always trend in the right direction. As mentioned above, total spiking activity is only one measure of spiking. It is possible that these drugs alter the pattern of spiking that translates into an altered calcium transients which may be important for triggering the plasticity. Further, we acknowledge that we cannot rule out a role for NMDARs contributing to GABAergic scaling (3rd and 5th paragraph of discussion). Based on the variability that we observe and the nature of our MEA recordings we cannot precisely determine how the total activity or pattern of activity changes with drug application in the specific cells that we target for whole cell recordings, and this is now discussed (final paragraph of discussion). Again, it is important to note that we are going a step beyond most homeostatic plasticity studies that add a drug and simply assume it is having an effect on spiking (e.g. CNQX was initially thought to completely abolish spiking, but clearly does not). However, we believe that the most parsimonious explanation of our results supports our proposal that GABAergic scaling is a strong candidate as a spike rate homeostat. Regardless, in the resubmission we have included a broader discussion about these possibilities, and recognize that we cannot rule out the possibility that AMPAergic transmission could contribute to upward GABAergic scaling (2nd paragraph discussion).

The paper also suggests that the timing of the GABA changes coincides with the spiking changes, but while they have the time course of the spiking changes and recovery, they only have the 24h time point for synaptic changes. It is impossible to conclude how the time courses align without more data.

We can only say that by the 24 hour CNQX time point, when overall spiking is recovered in some but not all cultures and bursts have not recovered, that GABAergic scaling has already occurred. We now state this more clearly in the resubmission (near the end of the 2nd paragraph of the discussion).

Reviewer #3 (Recommendations For The Authors):

The statistics are inadequately described. The full information including actual p values should be given, particularly for the non-significant trends reported.

We have done this in Figure legends.

The abstract and introduction give the impression that GABA and Glu HSP are independent, though most work links them as occurring simultaneously and in a coordinated fashion to achieve homeostasis.

While it is true that many studies have triggered both forms of scaling with activity or transmission blockade, these studies have not addressed whether these forms of scaling are actually triggered in the same way mechanistically, except potentially for the one study that we mentioned (Joseph et al.,). Our results suggest they are independent. We now do mention the idea that these two forms of scaling have been assumed to be commonly triggered (3rd paragraph introduction).

The data in Fig 6 is presented as if BIC treatment is a novel result, although BIC/Gabazine/PTX have been used to induce down-scaling in many previous papers. While it's good to have the results, they should be put in proper context. As suggested in the paper, testing if decreased GABAR function would lead to upscaling does not make sense given all the previous data.

Figure 6 shows GABAergic upscaling in response to GABAR block (bicuculline), but we are aware of only two other studies that looked at GABAergic scaling after treating with a GABAR blocker and they found upscaling but this was in hippocampal cultures, not cortical cultures (Peng et al., 2010 - PMID: 21123568, Pribiag et al., 2014 - PMID: 24753587). We now mention this in the results section describing Figure 6. While many studies have blocked GABARs and find AMPAergic downscaling, we are addressing the triggers for GABAergic scaling in Figure 6.

Is Fig S4B mislabeled? The title says spike rate but the graph axis says burst frequency.

The reviewer is correct and we have now adjusted this.

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